# Experiment 045: Predicted Trigenic Profiles

Every figure to date, complete and partial

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October 28, 2026

Supplementary Fig. 21 of the manuscript (Figure 1) is composed from arm R2 and is complete for what it uses: the Tier A seed mean of three checkpoints, a predicted trigenic interaction $\tau$ for each of the 6,607 genes against each of the 420 query pairs of Kuzmin et al. 2018 (doi:10.1126/science.aao1729). Its two data panels are drawn again for each of the four arms (Figures 2 to 5). The leave-one-gene-out threshold sweeps behind the PCC $> 0.6$ rule follow for each arm (Figures 6 to 9), then a sanity figure for one checkpoint (Figure 10). Every figure is built from Tier A. The Tier B core is scored and pulled, the Tier B extension ran to the workers' time limit, and no Tier B analysis exists yet (Section 2). Each caption ends with a status sentence; terms are defined in Table 1.

*Section status:* ✓ ready ■ author-reviewed, keep stable ✗ not done, agents may edit freely

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# 1 Controlled vocabulary ✗

**Table 1:** Terms used in the captions and tables, in alphabetical order.

| Term | Definition |
|---|---|
| $\|\tau\| > 0.08$ | Kuzmin's magnitude rule for a trigenic interaction, applied here to measured and predicted $\tau$ alike. The measured call also requires $p < 0.05$, which a prediction does not carry. |
| Arm | One predictor family, three seeds each, none told which screen a cell comes from (no dataset-name token). R1: the 010 CGT, random split. R2: the 025 closure CGT, random split. Q1 and Q2: CGTs trained on a query-pair-disjoint split, which holds out 89 of the 420 query pairs (43 validation, 46 test); Q2 adds a fitness co-head. |
| Base rate | The leave-one-gene-out precision of a term drawn at random: the mean share of the gold terms an annotated gene bears. |
| Degenerate profile | A profile with PCC above 0.99 to 10 or more others after preprocessing: the model's single output for genes it has no information about. Counted in coverage, excluded from the network. |
| Dense gene | A gene with 50 or more measured array-role cells; 1,182 genes. |
| Density-matched threshold | The predicted-profile PCC at which the dense genes' predicted network has as many edges as their measured network at 0.6. |
| Gate | Re-scoring a checkpoint on its own validation cells and comparing the Pearson with the value W&B logged in training. A pass is within 0.002; a soft pass is scored and flagged, within a hard tolerance of 0.05. |
| Gold standard | SGD GO Biological Process annotations from the frozen GAF (2025-03-18), evidence codes EXP, HDA, HEP, HMP, HTP, IDA, IEP, IMP, IPI only (HGI, IGI are genetic-interaction inferences and would be circular), propagated through `is_a`, terms with 10 to 500 bearers: 1,291 terms on 4,393 genes. |
| LOGO precision | Leave-one-gene-out precision: each annotated gene's own terms are hidden, terms are transferred to it from its neighbors, and precision is the share of transferred terms the gene bears in the gold standard. A lower bound, since SGD curation is incomplete. |
| Most-specific term | A transferred term with no transferred descendant in the same gene's called set. |
| Profile | A gene's $\tau$ against each of the 420 query pairs, the gene in the array role. Before correlation each query pair's mean over all genes is removed (pair-centering) and the first principal component is removed. |
| Prospective genes | The 16 uncharacterized genes that gained an experimental GO BP term between the frozen GAF and the SGD GAF of 2026-10-20; kept out of the candidates as a prospective test. |
| Query pair | One of the 420 double-mutant queries of the Kuzmin 2018 trigenic screen. |
| Seed mean | The mean over an arm's three best checkpoints of the Tier A predictions. |
| Supporting neighbors ($k$) | The linked neighbors of a gene that bear the transferred term. |
| Tier A | Every gene against every measured query pair: 2,774,100 triples per checkpoint (6,607 × 420 cells less the 840 where the array gene is one of the pair's genes). |
| Tier B | New query pairs among a 165-gene panel (13,522 pairs), every gene scored against them: a core of 19 blocks for each of the 12 best checkpoints and an extension of further pairs for 8 of them (Table 2). |
| Transfer | For one gene, a hypergeometric test of each gold term among its linked neighbors (profile PCC > 0.6) against the annotated population; Benjamini-Hochberg $q$ over every (gene, term) test of the evaluation set, called at $q < 0.05$ with at least 2 supporting neighbors. |
| Uncharacterized gene | One of the 668 genes of the experiment 013 list: 405 with a Kuzmin record, 263 without. |
| Validation candidate | A non-essential uncharacterized gene with a transfer and no experimental GO BP annotation (EXP, HDA, HEP, HMP, HTP, IDA, IEP, IMP, IPI, HGI, IGI) in the frozen GAF or in the SGD GAF of 2026-10-20, the prospective genes excluded. |

# 2 Status of the scoring and the analysis ✗

**Scoring.** Phases P0 to P4 are complete and pulled to GilaHyper (Table 2). P5, the Tier B extension, ran on IGB until the five-day limit of workers 2431557 and 2431558, 2026-10-28 11:30; 185 of its 208 items are on GilaHyper, the newest block written at 2026-10-28 11:07. The items that had not run by the limit were cut off, which the queue was designed to accept (plan G9).

**Table 2:** The scoring queue by phase. Planned items are the rows of `results/queue_plan.csv`; an item counts as pulled when its gate report or block manifest is present in `$DATA_ROOT/045/` on GilaHyper, so the pulled count can lag the workers on IGB. One item is one checkpoint file scored on one index block of at most 2 million triples.

| Phase | Items | Planned | Pulled | Triples planned (millions) |
|---|---|---|---|---|
| P0 | gates, each file's own validation set, best and last files: R2 × 3, R1 × 3, Q1 × 6, Q2 × 6 | 18 | 18 | 0.7 |
| P1 | benchmark block, best and last files: R2 × 3, R1 × 3, Q1 × 6, Q2 × 6; 1 block per file | 18 | 18 | 8.9 |
| P2 | Tier A, best files: R2 × 3, R1 × 3, Q1 × 3, Q2 × 3; 2 blocks per file | 24 | 24 | 33.3 |
| P3 | Tier B core, best files: R2 × 3, R1 × 3, Q1 × 3, Q2 × 3; 19 blocks per file | 228 | 228 | 447.9 |
| P4 | Tier A, last files: Q1 × 3, Q2 × 3; 2 blocks per file | 12 | 12 | 16.6 |
| P5 | Tier B extension, best files: R2 × 3, R1 × 3, Q1 × 1, Q2 × 1; 26 blocks per file | 208 | 185 | 415.9 |

**Analysis.** Every figure is built from Tier A, and every data panel uses the seed mean of an arm's three best checkpoints. No Tier B analysis has been written. The plan (G9) demotes Tier B to a descriptive supplementary analysis with a cross-assignment control, because its panel was built around the term bearers of the validation genes. The benchmark blocks (P1) and the Q last files (P4) are scored and pulled, and no figure here uses them; nor is there a figure for the plan's held-out benchmark, masking test, residualization or calibration by tier (F7 items 1 to 4).

**Gates.** The 3 R1 checkpoints re-score below the validation Pearson W&B logged for them (Table 3); for `lzs9pcj3` the re-score is 0.420 against 0.452, the same at bf16 and fp32 and on a code tree from the week it was trained (plan G9). They pass under the hard tolerance of 0.05 and are scored with a flag; the cause of the gap is unexplained. Every R2 and Q gate passes.

**Table 3:** The 12 best checkpoints. Gate: Pearson of the re-scored predictions on the file's own validation cells against the value W&B logged at training time; a pass is within 0.002, a soft pass is scored and flagged. Precision: the scoring precision recorded in the block manifests. Tier A and Tier B: blocks pulled to GilaHyper (Tier A is 2 blocks per file, the Tier B core 19). The last three columns are Pearson correlations from `inspect_scores_045.py`: prediction against measured $\tau$ on the arm's test cells; the same on validation cells after removing each query pair's mean (within-pair); and predicted against measured profile PCC over dense gene pairs, the prediction restricted to the measured cells (profile).

| Arm | Run | Precision | Gate | W&B | Verdict | Tier A | Tier B | Test | Within-pair | Profile |
|---|---|---|---|---|---|---|---|---|---|---|
| R2 | `8fuwsk48` | bf16 | 0.501 | 0.501 | pass | 2 | 44 | 0.487 | 0.360 | 0.547 |
| R2 | `srl1xzdf` | bf16 | 0.499 | 0.499 | pass | 2 | 41 | 0.482 | 0.360 | 0.546 |
| R2 | `yb4gjh51` | bf16 | 0.498 | 0.498 | pass | 2 | 41 | 0.481 | 0.346 | 0.458 |
| R1 | `c7671wgj` | bf16 | 0.438 | 0.462 | soft pass | 2 | 43 | 0.427 | 0.274 | 0.251 |
| R1 | `lzs9pcj3` | bf16 | 0.420 | 0.452 | soft pass | 2 | 43 | 0.409 | 0.258 | 0.265 |
| R1 | `yv4r30bi` | bf16 | 0.416 | 0.447 | soft pass | 2 | 41 | 0.400 | 0.255 | 0.305 |
| Q1 | `e9gpr997` | fp32 | 0.272 | 0.270 | pass | 2 | 19 | 0.132 | 0.017 | 0.066 |
| Q1 | `if1gdrya` | fp32 | 0.276 | 0.275 | pass | 2 | 43 | 0.232 | 0.085 | 0.167 |
| Q1 | `s1vx2zgw` | fp32 | 0.261 | 0.262 | pass | 2 | 19 | 0.168 | 0.018 | 0.070 |
| Q2 | `baozg4ou` | fp32 | 0.242 | 0.243 | pass | 2 | 19 | 0.157 | 0.025 | 0.080 |
| Q2 | `s4buqxee` | fp32 | 0.255 | 0.254 | pass | 2 | 19 | 0.094 | 0.024 | 0.059 |
| Q2 | `wpr18i1o` | fp32 | 0.279 | 0.278 | pass | 2 | 41 | 0.195 | 0.095 | 0.224 |

# 3 Figures ✗

Figure 1 is the manuscript figure. Figures 2 to 5 draw its data panels for each arm, Figures 6 to 9 the threshold sweeps for each arm, and Figure 10 the score sanity check. Colors carry the same meaning throughout: gray is measured by the screen, orange is predicted, and within a two-level bar the pale fill counts every cell while the dark fill counts the cells beyond $|\tau| > 0.08$.

SRC: uncharacterized-genes-profile-enricment.gen.py; paper/nature-biotech/figures/

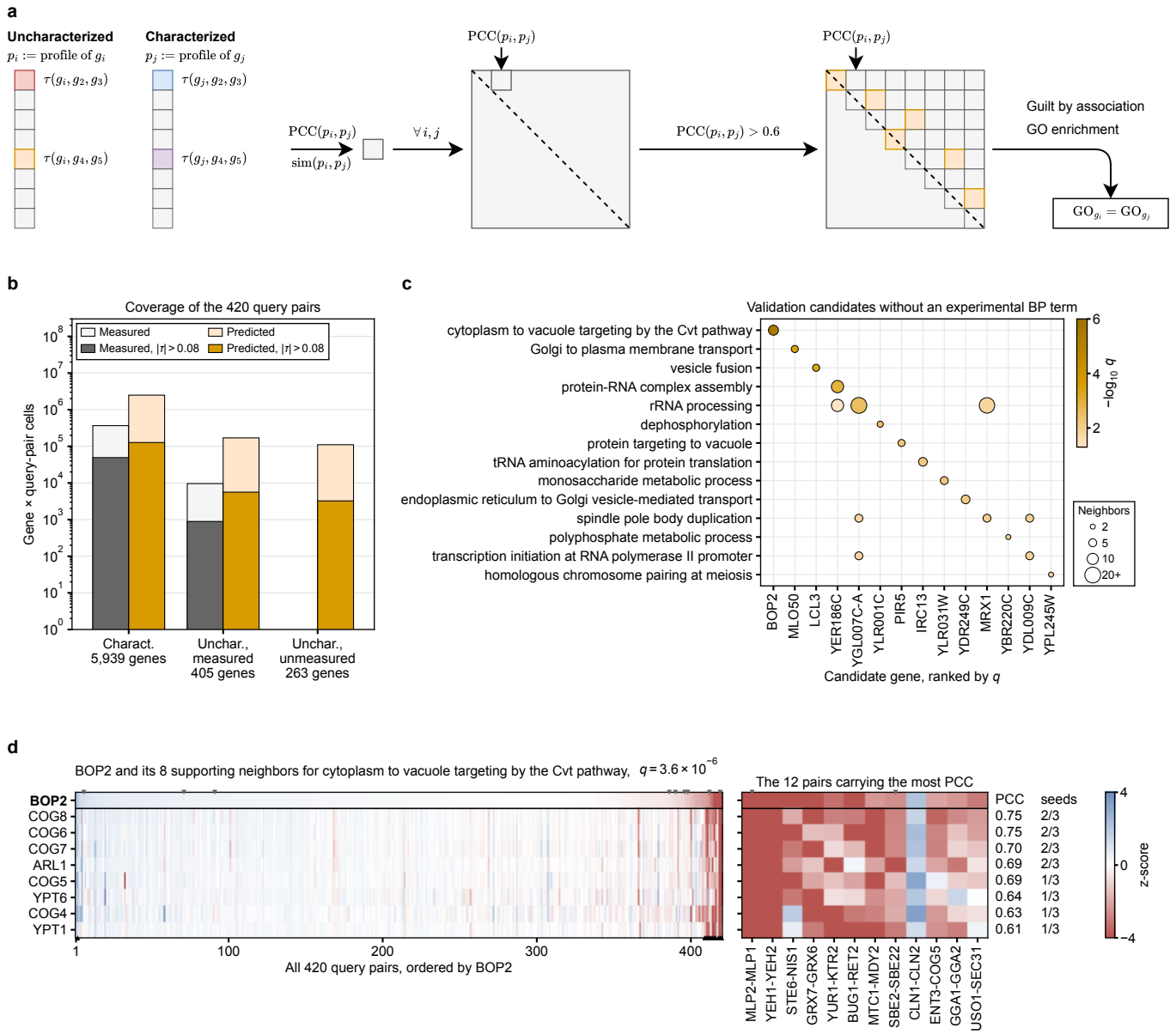



**Figure 1: Supplementary Fig. 21 as composed for the manuscript: predicted triple-interaction profiles transfer GO terms to uncharacterized genes by guilt by association (arm R2). (a)** The transfer rule: a gene's profile of predicted $\tau$ over the 420 query pairs is correlated with every other gene's by Pearson correlation (PCC), pairs with PCC $> 0.6$ are linked (orange cells of the thresholded matrix), and GO BP enrichment over a gene's linked neighbors transfers terms to it. **(b)** Gene × query-pair cells measured (gray) and predicted (orange) for 5,939 characterized genes, 405 uncharacterized genes the screen measured and 263 it never measured; the dark fill counts cells with $|\tau| > 0.08$. **(c)** The 14 top-ranked of 93 validation candidates, each with its best most-specific term, ranked by $q$; dot size is the number of supporting neighbors and color is $-\log_{10} q$, pale amber to bronze. **(d)** The evidence behind the top candidate (Figure 11): the centered profiles of *BOP2* and its supporting neighbors over all 420 query pairs, ordered by *BOP2*'s value, and the twelve pairs that carry the most of the correlation, with each neighbor's PCC and the number of single seeds in which it is a neighbor; red is negative, blue positive. **Status: complete for the Tier A seed mean of R2** (`8fuwsk48`, `srl1xzdf` and `yb4gjh51`); 416 degenerate profiles excluded, LOGO precision 0.147 against a base rate of 0.013 at PCC $> 0.6$. No Tier B data. The manuscript copy is marked `tent`; panel d and its caption sentence were added on 2026-10-28 at the owner's request, and the figure here is a byte copy of its PDF, 180 × 162 mm.

SRC: `profile_enrichment_panels_045.py --arm R2`

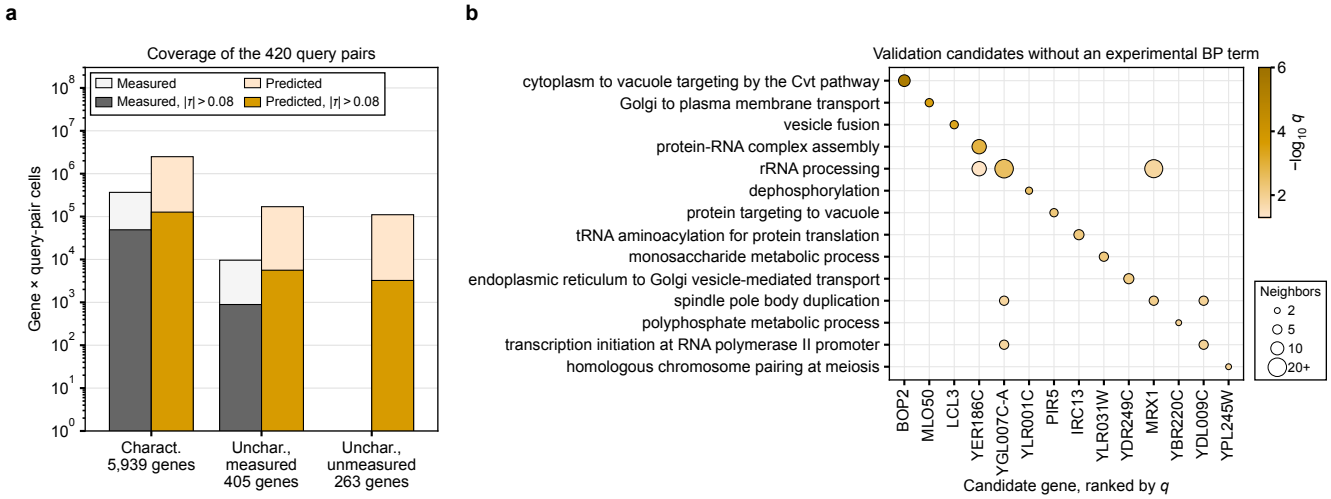



**Figure 2: Panels b and c of Supplementary Fig. 21 as standalone files, arm R2. (a)** Coverage of the 420 query pairs: the screen measured 376,732 cells, 13.3% of them beyond $|\tau| > 0.08$; R2 predicts all 2,774,100, 4.9% beyond the rule (5.4% on the measured cells). **(b)** The 14 top-ranked of 93 validation candidates (8 of them without a Kuzmin record); the first is *BOP2*, cytoplasm to vacuole targeting by the Cvt pathway ($k = 8$, $q = 3.6 \times 10^{-6}$). A gene column is at least as wide as the largest dot it holds plus 0.35 mm, so no two dots touch; an arm whose term names leave less width shows fewer genes. **Status: complete for the Tier A seed mean of R2** (`8fuwsk48`, `srl1xzdf` and `yb4gjh51`); these are the files embedded in Figure 1.

SRC: `profile_enrichment_panels_045.py --arm R1`

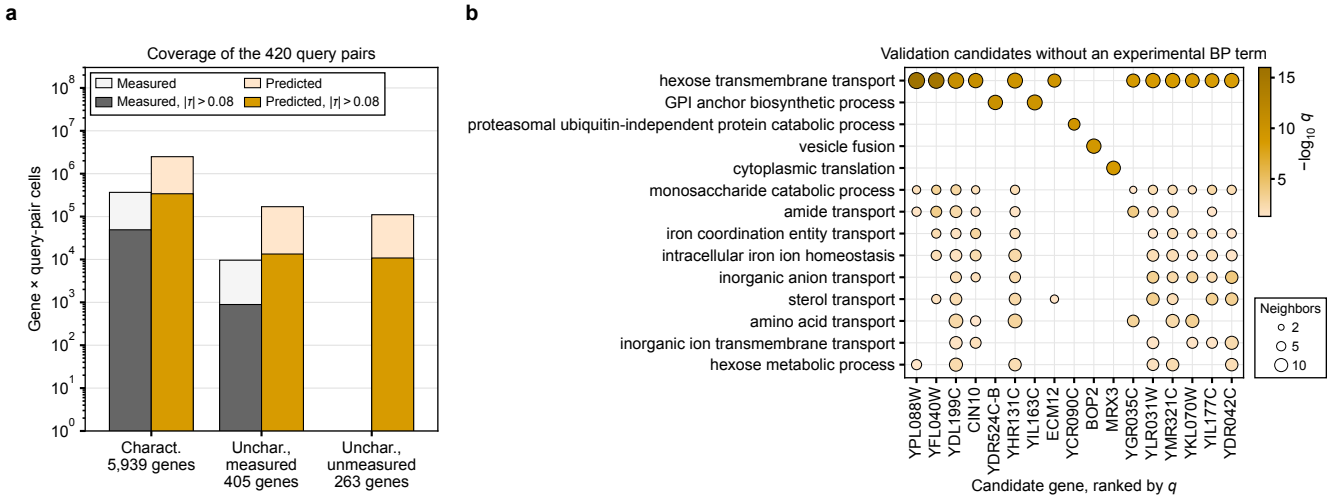



**Figure 3: The same two panels from arm R1, the 010 CGT. (a)** Coverage: R1 predicts 13.2% of its cells beyond $|\tau| > 0.08$ (13.9% on the measured cells) against 13.3% measured; only 10 of its profiles are degenerate, against 416 for R2. **(b)** The 17 top-ranked of 432 validation candidates (165 without a Kuzmin record); the first is *YPL088W*, hexose transmembrane transport ($k = 15$, $q = 1.6 \times 10^{-16}$). The panel stops at 17 genes, under the cap of 18, because its longest term names leave too little width for more gene columns at the spacing of Figure 2. At PCC $> 0.6$ R1 links 274,955 gene pairs against 86,794 for R2, and transfers terms to 462 of the 665 non-essential uncharacterized genes outside its degenerate set, against 107 of 608 for R2. **Status: complete for the Tier A seed mean of R1** (`c7671wgj`, `lzs9pcj3` and `yv4r30bi`), with the soft-pass gate flag every R1 checkpoint carries (Section 2). Not in the manuscript.

SRC: `profile_enrichment_panels_045.py --arm Q1`

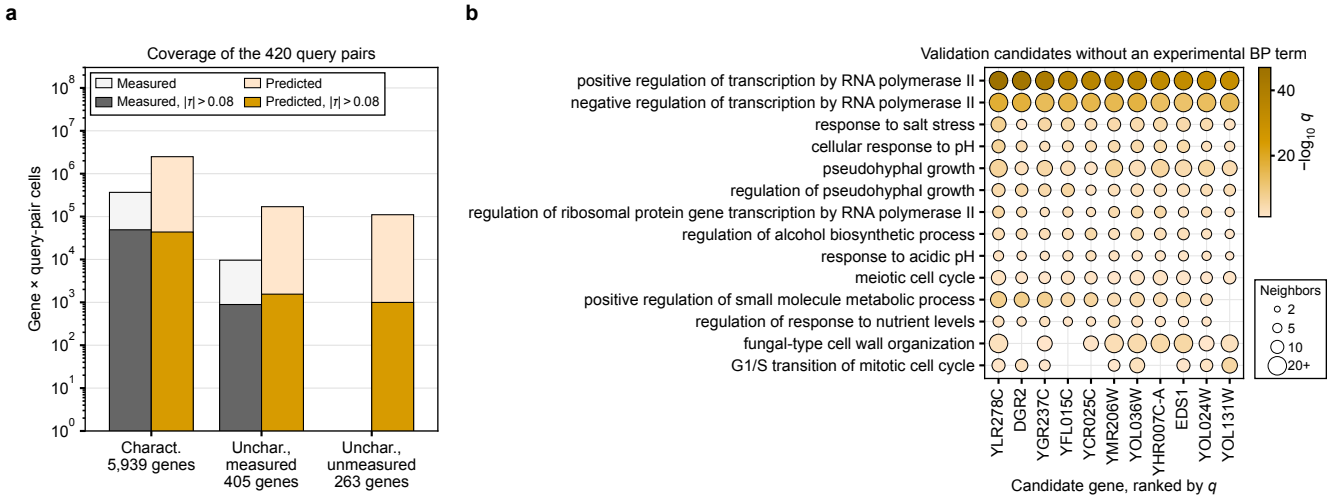



**Figure 4: The same two panels from arm Q1, trained on the query-pair-disjoint split. (a)** Coverage: Q1 predicts 1.7% of its cells beyond $|\tau| > 0.08$ (1.4% on the measured cells) against 13.3% measured, and 89 of its 420 columns are query pairs it never trained on. **(b)** The 11 top-ranked of 599 validation candidates (246 without a Kuzmin record); the first is *YLR278C*, positive regulation of transcription by RNA polymerase II ($k = 106$, $q = 1.2 \times 10^{-47}$). The panel stops at 11 genes, under the cap of 18, because its longest term names leave too little width for more gene columns at the spacing of Figure 2. Q1 links 1,521,596 gene pairs at PCC $> 0.6$, against 86,794 for R2. **Status: complete for the Tier A seed mean of Q1** (`e9gpr997`, `if1gdrya` and `s1vx2zgw`). Not in the manuscript.

SRC: `profile_enrichment_panels_045.py --arm Q2`

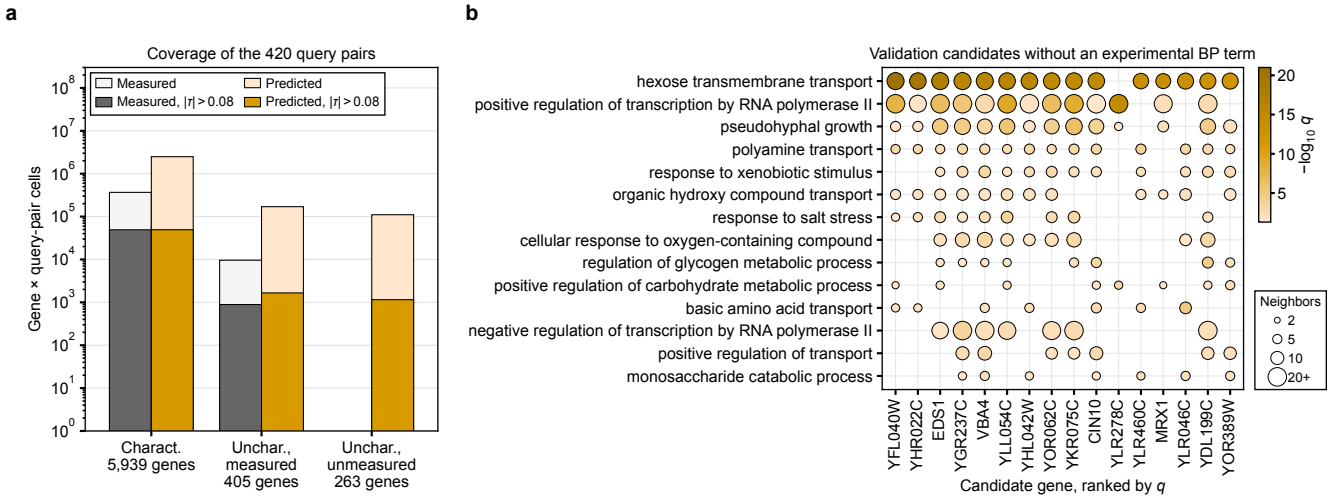



**Figure 5: The same two panels from arm Q2, trained on the query-pair-disjoint split with a fitness co-head. (a)** Coverage: Q2 predicts 1.9% of its cells beyond $|\tau| > 0.08$ (1.7% on the measured cells) against 13.3% measured, and 89 of its 420 columns are query pairs it never trained on. **(b)** The 16 top-ranked of 575 validation candidates (238 without a Kuzmin record); the first is *YFL040W*, hexose transmembrane transport ($k = 17$, $q = 1.6 \times 10^{-21}$). The panel stops at 16 genes, under the cap of 18, because its longest term names leave too little width for more gene columns at the spacing of Figure 2. Q2 links 603,188 gene pairs at PCC $> 0.6$, against 86,794 for R2. **Status: complete for the Tier A seed mean of Q2** (`baozg4ou`, `s4buqxee` and `wpr18i1o`). Not in the manuscript.

SRC: `profile_enrichment_panels_045.py --arm R2, sweep_figure`

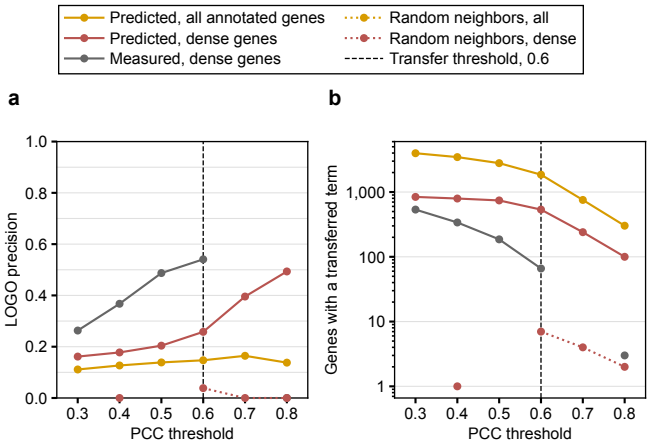



**Figure 6: The PCC threshold against leave-one-gene-out precision, arm R2.** **(a)** LOGO precision of the transferred terms. **(b)** Annotated genes with a transferred term (log scale). The dashed line is the threshold of the transfer panels, 0.6. Orange: predicted profiles over all 4,393 annotated genes; red: predicted profiles over the 1,042 dense annotated genes; gray: measured profiles over the same dense genes; dotted: size-matched random neighbors, which make at most 26 calls at any threshold. At 0.6 predicted profiles reach precision 0.147 on 1,855 genes against a base rate of 0.013; on the dense genes, 0.258 on 536 genes for predicted against 0.541 on 66 for measured profiles, and the predicted dense network recovers 382 of the 546 measured edges. The density-matched threshold is 0.84. A point is missing where no term was called. **Status: complete for the Tier A seed mean of R2**; a supporting figure, not a manuscript panel.

SRC: `profile_enrichment_panels_045.py --arm R1, sweep_figure`

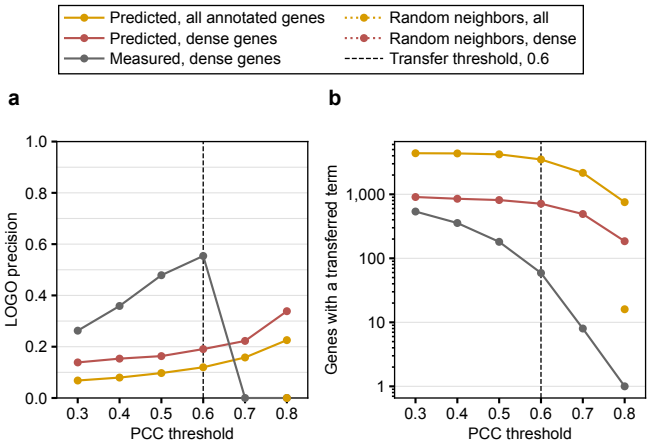



**Figure 7: The same sweep for arm R1.** Lines and colors as in Figure 6. At 0.6 predicted profiles reach precision 0.120 on 3,502 genes against a base rate of 0.013; on the dense genes, 0.191 on 713 genes for predicted against 0.554 on 59 for measured profiles, and the predicted dense network recovers 189 of the 352 measured edges. The most calls random neighbors make at any threshold is 45. The density-matched threshold is 0.88. **Status: complete for the Tier A seed mean of R1**, with the R1 gate flag; drawn by the same run as Figure 3.

SRC: `profile_enrichment_panels_045.py --arm Q1, sweep_figure`

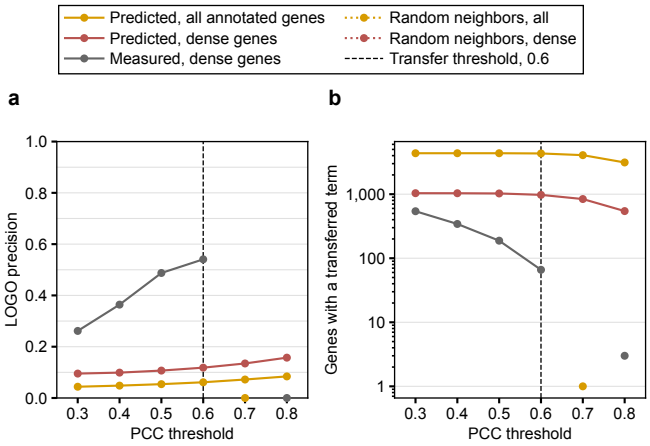



**Figure 8: The same sweep for arm Q1.** Lines and colors as in Figure 6. At 0.6 predicted profiles reach precision 0.061 on 4,325 genes against a base rate of 0.013; on the dense genes, 0.118 on 976 genes for predicted against 0.541 on 66 for measured profiles, and the predicted dense network recovers 173 of the 541 measured edges. The most calls random neighbors make at any threshold is 1. The density-matched threshold is 0.93. **Status: complete for the Tier A seed mean of Q1**; a supporting figure, not a manuscript panel.

SRC: `profile_enrichment_panels_045.py --arm Q2, sweep_figure`

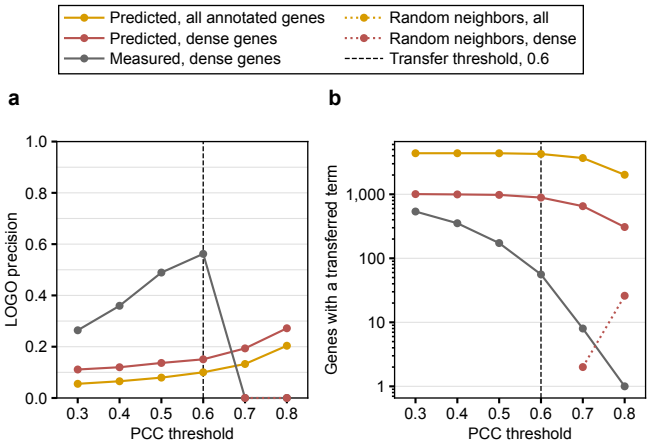



**Figure 9: The same sweep for arm Q2.** Lines and colors as in Figure 6. At 0.6 predicted profiles reach precision 0.100 on 4,264 genes against a base rate of 0.013; on the dense genes, 0.151 on 889 genes for predicted against 0.562 on 56 for measured profiles, and the predicted dense network recovers 147 of the 349 measured edges. The most calls random neighbors make at any threshold is 72. The density-matched threshold is 0.88. **Status: complete for the Tier A seed mean of Q2**; a supporting figure, not a manuscript panel.

SRC: `inspect_scores_045.py`

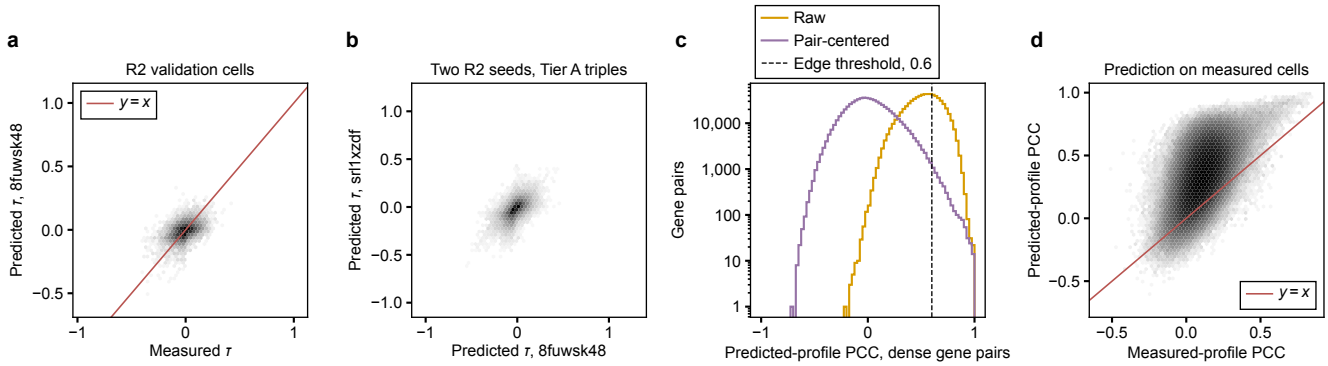



**Figure 10: Score sanity for one checkpoint, R2 `8fuwsk48`: a single file, not a seed mean.** Gray hexagons are counts on a log scale and the red line is $y = x$. **(a)** Predicted against measured $\tau$ on the 37,673 validation cells, Pearson $r = 0.501$. **(b)** The predictions of `8fuwsk48` against those of `srl1xzdf` on a random subsample of the Tier A triples; over all of them $r = 0.703$. **(c)** Predicted-profile PCC over the dense gene pairs, raw (orange) and pair-centered (purple), dashed line at 0.6: over all gene pairs 9,300,174 raw PCCs exceed 0.6 (median 0.57) and 215,763 after pair-centering, with no principal component removed. **(d)** Predicted-profile PCC, the prediction restricted to the measured cells, against measured-profile PCC over 697,827 dense gene pairs, $r = 0.547$. **Status: partial.** The statistics exist for all 12 best checkpoints (Table 3, from `results/score_inspection_per_file.csv`), but the script draws this figure for its reference checkpoint only, the first R2 file.

# 4 Candidate dossiers ✗

The 14 genes drawn in Supplementary Fig. 21c are opened up one by one here, from the same R2 network the panel uses: the seed mean of the three R2 Tier A files, pair-centered, first principal component removed, the 416 degenerate profiles left out. The network was rebuilt from the score files, and its 374 transfers to the 93 validation candidates equal the panel script's term for term; the 16 prospective genes, whose outcome is known, were run through the same dossier for comparison. Each dossier gives the transferred term and the neighbors it rests on, the query pairs that carry those correlations, how the call fares in each single checkpoint, in the other three arms and without the dominant query pairs, whether the model's own inputs already imply it, and every GO annotation SGD added for the gene after the frozen GAF of 2025-03-18. Table 4 defines the terms the dossiers add to Table 1; Section 5 reports each route over all 93 candidates. Per-gene tables for all 93 candidates and the 16 prospective genes, one fact per row, are in `experiments/045-trigenic-profile-inference/results/profile_enrichment/R2/dossiers/`, with one JSON file per gene under `genes/`.

**Table 4:** Terms the dossiers add, in alphabetical order.

| Term | Definition |
|---|---|
| AUC against decoys | For a candidate and its best term, the area under the ROC curve with which the candidate's PCC row ranks the term's bearers above the other annotated genes (0.5 is no separation), set against the same AUC for 100 decoys: annotated genes that do not bear the term, nearest to the candidate in Tier A degree. In Tier A the neighbors chose the term, so only the Tier B AUC is evidence the call did not select. |
| Column dropout | The network and the transfer rebuilt with 10 of the 420 query pairs removed: the 10 whose pair-centered predictions spread widest over genes, or 10 drawn at random (5 draws). A call that survives random drops but not the widest-spread drop rests on those few pairs. |
| Driving pair | A query pair that carries much of the PCC between a candidate and its supporting neighbors. With each profile standardized to $z$, the PCC is the mean over pairs of $z_{\text{gene}}\, z_{\text{neighbor}}$, so each pair's product is its exact share. |
| Embedding guilt by association | The panel's transfer rule applied to each gene's $k$ most cosine-similar genes in the learned gene embedding, $k$ equal to its profile degree at PCC $> 0.6$. |
| Input graphs | The nine gene graphs the R2 model reads as input (physical, regulatory, TFLink and the six STRING 12.0 channels (neighborhood, fusion, co-occurrence, co-expression, experimental, database)), taken from the cell graph stored in checkpoint `8fuwsk48`; the three R2 configurations list the same nine. Their union links 1,514,071 gene pairs, median degree 345. |
| Learned gene embedding | `H_genes`, the 180-dimensional wildtype gene representation an R2 checkpoint's encoder hands to its perturbation head, written out by each checkpoint's gate run; cosine similarity is averaged over the three seeds. |
| New annotation | A (gene, GO term, evidence code) in the SGD GAF of 2026-10-20 that the frozen GAF (2025-03-18) does not hold, any aspect and any code, rows qualified NOT excluded, alternative GO ids resolved to their primary term. Rows coded ND (no biological data) are counted apart. |
| Seed-mean degree | The number of genes linked to a gene at PCC $> 0.6$ in the seed-mean network, the $k$ of top-$k$ recovery. |
| Share of the top two pairs | The fraction of a supporting neighbor's PCC carried by its two largest pair products, averaged over the supporting neighbors of the best term. |
| Single-seed network | The panel's rule applied to one R2 checkpoint's Tier A file alone: the same preprocessing, that file's own degenerate set and its own Benjamini-Hochberg correction over the ring 2 tests. |
| Tier B core columns | The 5,651 new query pairs among the Tier B panel that hold a prospective gene or a dense ME-relevant gene, every gene scored against each by the three R2 checkpoints. |
| Top-$k$ recovery | A supporting neighbor is kept in a seed when it is among that seed's $k$ most correlated genes, $k$ the seed-mean degree. A single seed's PCCs run lower than the mean's, so a fixed threshold of 0.6 drops most links in every seed. |
| Within two steps | The ring 3 test: a term within two `is_a` steps up or down (parents, grandparents, children, grandchildren; no siblings). The broader variant also follows `part_of`, the only edge by which a function or component term can reach a process term. |

## 4.1 The drawn candidates at a glance ✗

**Table 5:** The 14 drawn candidates. Description is the first clause of SGD's (verbatim); best term is the most specific transferred term with the lowest $q$; $k$ supporting neighbors. Seeds: single-seed networks (of three) that call the term. Arms: other arms whose own transfer gives the term or one within two `is_a` steps. New: evidence codes of new annotations since 2025-03-18 (none is within two steps of a transferred term). Tier B: in the 165-gene panel. Graph and Emb.: union-graph and embedding guilt by association call the term. Top 2: share of the supporting PCC carried by the two strongest pairs.

| # | Gene | SGD description | Best term | $q$ | $k$ | Seeds | Arms | New | Tier B | Graph | Emb. | Top 2 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | *BOP2* | Protein of unknown function | cytoplasm to vacuole targeting by the Cvt pathway | $3.6 \times 10^{-6}$ | 8 | 2/3 | R1 | – | no | no | no | 0.53 |
| 2 | *MLO50* | Mitochondrial protein of unknown function | Golgi to plasma membrane transport | $3.2 \times 10^{-4}$ | 4 | 1/3 | R1 | – | no | no | no | 0.50 |
| 3 | *LCL3* | Protein of unknown function | vesicle fusion | $4.7 \times 10^{-4}$ | 4 | 0/3 | – | IBA | no | no | no | 0.37 |
| 4 | *YER186C* | Protein of unknown function | protein-RNA complex assembly | $1.2 \times 10^{-3}$ | 12 | 0/3 | – | – | no | no | no | 0.25 |
| 5 | *YGL007C-A* | Putative protein of unknown function | rRNA processing | $2.8 \times 10^{-3}$ | 23 | 2/3 | – | – | no | no | yes | 0.15 |
| 6 | *YLR001C* | Protein of unknown function | dephosphorylation | $3.1 \times 10^{-3}$ | 3 | 0/3 | Q1 | – | no | no | no | 0.21 |
| 7 | *PIR5* | Protein of unknown function | protein targeting to vacuole | $5.8 \times 10^{-3}$ | 4 | 0/3 | – | – | no | no | no | 0.38 |
| 8 | *IRC13* | Putative protein of unknown function | tRNA aminoacylation for protein translation | $6.1 \times 10^{-3}$ | 6 | 1/3 | – | – | no | no | no | 0.29 |
| 9 | *YLR031W* | Protein of unknown function | monosaccharide metabolic process | $6.3 \times 10^{-3}$ | 5 | 1/3 | R1 | – | yes | no | no | 0.18 |
| 10 | *YDR249C* | Protein of unknown function | endoplasmic reticulum to Golgi vesicle-mediated transport | $7.2 \times 10^{-3}$ | 6 | 1/3 | – | – | no | no | no | 0.44 |
| 11 | *MRX1* | Protein that associates with mitochondrial ribosome | spindle pole body duplication | $9.2 \times 10^{-3}$ | 5 | 0/3 | Q1 | – | no | no | yes | 0.14 |
| 12 | *YBR220C* | Protein of unknown function | polyphosphate metabolic process | $1.1 \times 10^{-2}$ | 2 | 0/3 | R1, Q1 | IBA | no | no | no | 0.27 |
| 13 | *YDL009C* | Protein of unknown function | transcription initiation at RNA polymerase II promoter | $1.2 \times 10^{-2}$ | 5 | 0/3 | – | – | no | no | no | 0.18 |
| 14 | *YPL245W* | Protein of unknown function | homologous chromosome pairing at meiosis | $1.2 \times 10^{-2}$ | 2 | 0/3 | Q1, Q2 | – | yes | no | yes | 0.24 |

## 4.2 The evidence behind the top three transfers ✗

Figures 11 to 13 show, for the three top-ranked candidates, the profiles the transfer compares and the query pairs that carry the comparison.

SRC: `candidate_dossiers_045.py, plot_evidence`

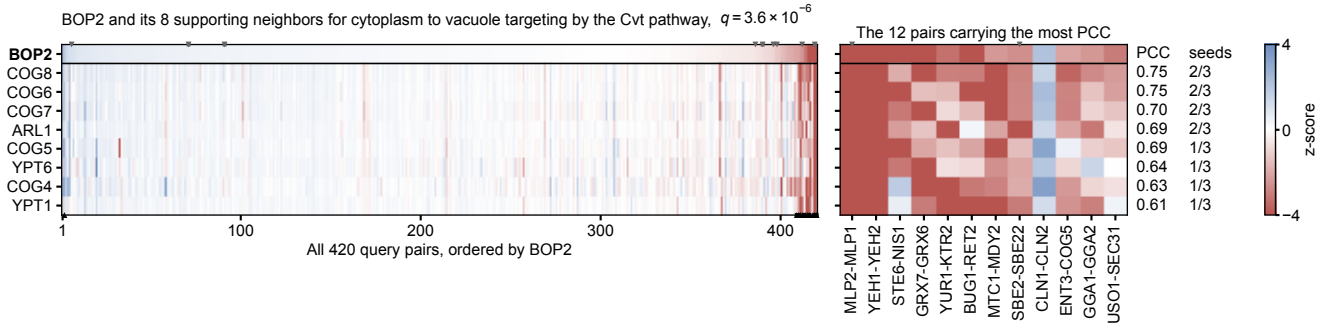



**Figure 11: The *BOP2* transfer rests on a few query pairs in which *BOP2* and its supporting neighbors share strong negative predicted interactions.** Each row is one gene's centered profile (pair-centered, first principal component removed) standardized to a $z$-score, the quantity the PCC averages; red is negative, blue positive, clipped at $\pm 4$. The top row is *BOP2*, the rows below its 8 supporting neighbors for cytoplasm to vacuole targeting by the Cvt pathway. **Left:** all 420 query pairs, ordered by *BOP2*'s value; gray triangles above mark the pairs where Kuzmin et al. measured *BOP2*, black triangles below the pairs drawn at right. **Right:** the twelve pairs with the largest mean product, named; the columns give each neighbor's PCC with *BOP2* and the number of single seeds in which it is among *BOP2*'s 48 most correlated genes. Two pairs carry 0.53 of each supporting PCC on average. **Status: complete for the Tier A seed mean of R2** (`8fuwsk48`, `srl1xzdf` and `yb4gjh51`); one candidate.

SRC: `candidate_dossiers_045.py, plot_evidence`

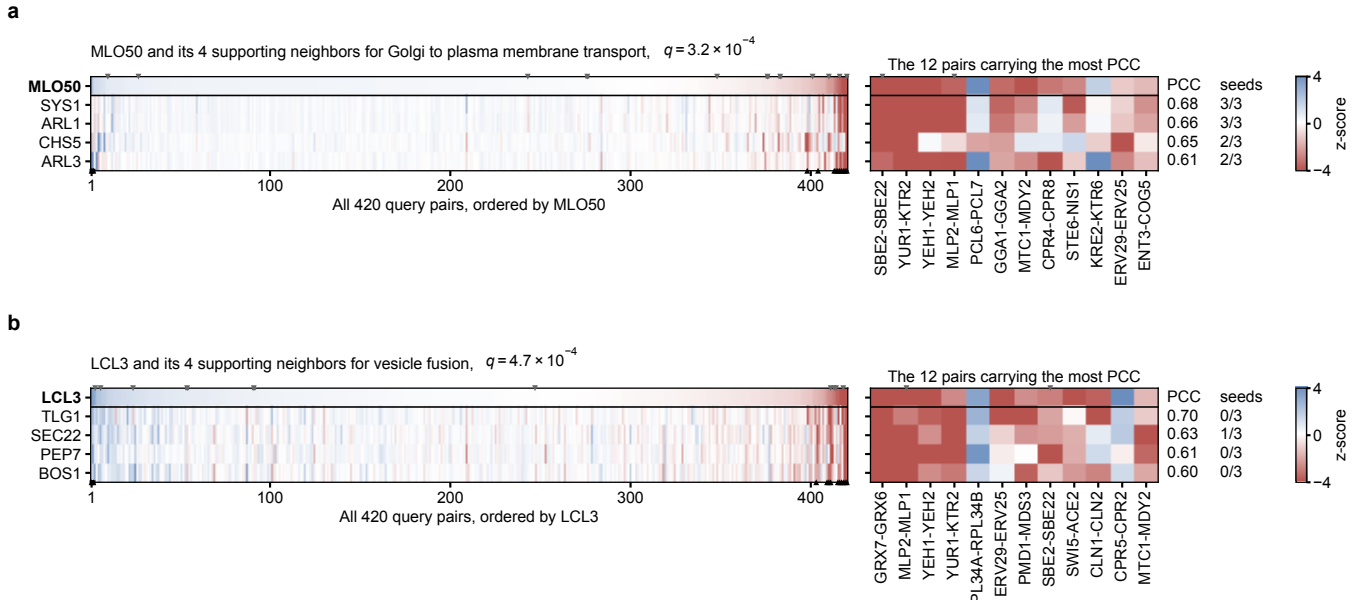



**Figure 12: The second and third candidates rest on the same few query pairs as the first.** Layout, color and columns as in Figure 11. **(a)** *MLO50* and its 4 supporting neighbors for Golgi to plasma membrane transport; two pairs carry 0.50 of each supporting PCC on average. **(b)** *LCL3* and its 4 supporting neighbors for vesicle fusion; two pairs carry 0.37 of each supporting PCC on average. **Status: complete for the Tier A seed mean of R2** (`8fuwsk48`, `srl1xzdf` and `yb4gjh51`); two candidates.

SRC: `candidate_dossiers_045.py, plot_evidence`

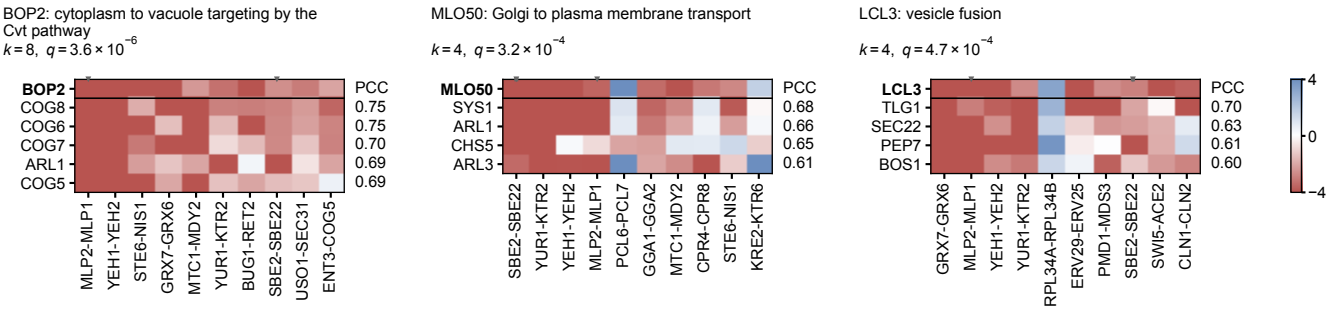



**Figure 13: The three top-ranked candidates side by side, each over the ten query pairs that carry its transfer.** Rows as in Figure 11, the gene and up to five supporting neighbors; columns are the gene's ten pairs with the largest mean product with its supporting neighbors, named. The same pairs recur across the three (GRX7-GRX6 and others; Section 5.3). Color as in Figure 11; gray triangles mark the gene's measured cells. **Status: complete for the Tier A seed mean of R2** (`8fuwsk48`, `srl1xzdf` and `yb4gjh51`); the candidate for a panel d of Supplementary Fig. 21 is Figure 11.

## 4.3 *BOP2* (YLR267W) ✗

**SGD.** Verified: "Protein of unknown function". Kuzmin et al. measured it in 9 array-role cells, 7 of them R2 training cells, and in 0 query-role cells.

**Prediction.** cytoplasm to vacuole targeting by the Cvt pathway (GO:0032258), $q = 3.6 \times 10^{-6}$: 8 of its 35 annotated neighbors bear the term against 0.39 expected by chance (49 bearers in the population); it has 48 neighbors in all and 25 transferred terms. GO defines the term: "A cytoplasm to vacuole targeting pathway that uses machinery common with autophagy."

**Supporting neighbors.** *COG8* (0.75), *COG6* (0.75), *COG7* (0.70), *ARL1* (0.69), *COG5* (0.69), *YPT6* (0.64), *COG4* (0.63), *YPT1* (0.61), with PCC. SGD describes the strongest three as *COG8*, "Component of the conserved oligomeric Golgi complex"; *COG6*, "Component of the conserved oligomeric Golgi complex"; *COG7*, "Component of the conserved oligomeric Golgi complex".

**What carries the correlation.** Two query pairs carry 0.53 of each supporting PCC on average; the three strongest are MLP2-MLP1, YEH1-YEH2, STE6-NIS1. Of the 18 cells at the two strongest pairs (the gene and its supporting neighbors), 15 were measured, 14 of them R2 training cells. Its most negative pair-centered predictions: STE6-NIS1 −0.437; MLP2-MLP1 −0.291, measured −0.260 (train); YEH1-YEH2 −0.251; its most positive: PCL6-PCL7 +0.117; CLN1-CLN2 +0.069.

**Robustness.** 2 of the three single-seed networks call the term (3 call it or a term within two steps); 0.00 of the supporting neighbors are kept in all three seeds by top-$k$ recovery; degenerate in seed: none. Other arms giving the same term: R1; the term or one within two steps: R1. The call survives the widest-spread column dropout: yes; random dropouts: 5 of 5.

**Baselines.** A share of 0.12 of the supporting neighbors are adjacent to it in an input graph. Union-graph guilt by association calls the term: no (within two steps: yes); embedding guilt by association: no (within two steps: no). Its nearest genes in the learned embedding: *YPR039W*, *FLO1*, *IRC16*, *YDR249C*, *ECM8*.

**New since 2025-03-18.** none.

**To look for.** Every supporting neighbor is a Golgi trafficking gene by SGD's description: five subunits of the conserved oligomeric Golgi complex, the soluble GTPase *ARL1* and the Rab GTPases *YPT6* and *YPT1*. The call has the lowest $q$ in the panel, and two single seeds make it, as they do for only *BOP2* and *YGL007C-A* among the drawn. Hypothesis (untested): *BOP2* acts in Golgi or Golgi-to-vacuole traffic with that module. Literature and phenotype data to check: published *BOP2* deletion phenotypes in Cvt or vacuolar sorting readouts (aminopeptidase I maturation, carboxypeptidase Y secretion), Golgi glycosylation, and any physical interaction with the COG subunits. Half of the signal sits in two query pairs, one of them a measured training cell, so a wet-lab test of the trigenic interactions at those two pairs would test the call directly.

## 4.4 *MLO50* (YLR049C) ✗

**SGD.** Verified: "Mitochondrial protein of unknown function". Kuzmin et al. measured it in 11 array-role cells, 9 of them R2 training cells, and in 0 query-role cells.

**Prediction.** Golgi to plasma membrane transport (`GO:0006893`), $q = 3.2 \times 10^{-4}$: 4 of its 14 annotated neighbors bear the term against 0.08 expected by chance (24 bearers in the population); it has 18 neighbors in all and 9 transferred terms. GO defines the term: "The directed movement of substances from the Golgi to the plasma membrane in transport vesicles that move from the trans-Golgi network to the plasma membrane, where they fuse and release their contents by exocytosis."

**Supporting neighbors.** *SYS1* (0.68), *ARL1* (0.66), *CHS5* (0.65), *ARL3* (0.61), with PCC. SGD describes the strongest three as *SYS1*, "Integral membrane protein of the Golgi"; *ARL1*, "Soluble GTPase with a role in regulation of membrane traffic"; *CHS5*, "Component of the exomer complex".

**What carries the correlation.** Two query pairs carry 0.50 of each supporting PCC on average; the three strongest are SBE2-SBE22, YUR1-KTR2, YEH1-YEH2. Of the 10 cells at the two strongest pairs (the gene and its supporting neighbors), 8 were measured, 7 of them R2 training cells. Its most negative pair-centered predictions: SBE2-SBE22 −0.777, measured −0.683 (train); YUR1-KTR2 −0.487; YEH1-YEH2 −0.364; its most positive: PCL6-PCL7 +0.261; KRE2-KTR6 +0.087.

**Robustness.** 1 of the three single-seed networks call the term (1 call it or a term within two steps); 0.50 of the supporting neighbors are kept in all three seeds by top-$k$ recovery; degenerate in seed: none. Other arms giving the same term: R1; the term or one within two steps: R1. The call survives the widest-spread column dropout: yes; random dropouts: 5 of 5.

**Baselines.** A share of 0.00 of the supporting neighbors are adjacent to it in an input graph. Union-graph guilt by association calls the term: no (within two steps: no); embedding guilt by association: no (within two steps: no). Its nearest genes in the learned embedding: *DFG5*, *EPO1*, *GID11*, *YBL062W*, *YGL024W*.

**New since 2025-03-18.** none.

**To look for.** The supporting neighbors are, in SGD's words, a Golgi membrane protein, two ARF-like GTPases and an exomer component, and *ARL1* supports *BOP2* as well. SGD calls *MLO50* mitochondrial. Hypothesis (untested): *MLO50* affects export from the late Golgi rather than a mitochondrial process. To check: the evidence behind SGD's mitochondrial assignment, and exomer-dependent export phenotypes of the deletion. The strongest pair is a measured training cell for *MLO50*, so part of the signal is the training data, not a prediction.

## 4.5 *LCL3* (YGL085W) ✗

**SGD.** Verified: "Protein of unknown function; mutant has long chronological lifespan; has homology to Staphylococcus aureus nuclease; GFP-fusion protein localizes to mitochondria; is induced in response to the DNA-damaging agent MMS". Kuzmin et al. measured it in 10 array-role cells, 8 of them R2 training cells, and in 0 query-role cells.

**Prediction.** vesicle fusion (GO:0006906), $q = 4.7 \times 10^{-4}$: 4 of its 13 annotated neighbors bear the term against 0.09 expected by chance (29 bearers in the population); it has 15 neighbors in all and 7 transferred terms. GO defines the term: "Fusion of the membrane of a transport vesicle with its target membrane."

**Supporting neighbors.** *TLG1* (0.69), *SEC22* (0.63), *PEP7* (0.61), *BOS1* (0.60), with PCC. SGD describes the strongest three as *TLG1*, "Essential t-SNARE that mediates fusion of vesicles with the late Golgi"; *SEC22*, "R-SNARE protein"; *PEP7*, "Adaptor protein involved in vesicle-mediated vacuolar protein sorting".

**What carries the correlation.** Two query pairs carry 0.37 of each supporting PCC on average; the three strongest are GRX7-GRX6, MLP2-MLP1, YEH1-YEH2. Of the 10 cells at the two strongest pairs (the gene and its supporting neighbors), 3 were measured, 2 of them R2 training cells. Its most negative pair-centered predictions: YEH1-YEH2 −0.114; GRX7-GRX6 −0.086; SWI5-ACE2 −0.086; its most positive: CPR5-CPR2 +0.088; STE6-NIS1 −0.038.

**Robustness.** 0 of the three single-seed networks call the term (0 call it or a term within two steps); 0.00 of the supporting neighbors are kept in all three seeds by top-$k$ recovery; degenerate in seed: none. Other arms giving the same term: none; the term or one within two steps: none. The call survives the widest-spread column dropout: no; random dropouts: 5 of 5.

**Baselines.** A share of 0.00 of the supporting neighbors are adjacent to it in an input graph. Union-graph guilt by association calls the term: no (within two steps: no); embedding guilt by association: no (within two steps: no). Its nearest genes in the learned embedding: *YNL089C*, *YJR038C*, *YDL012C*, *FUS1*, *YPR050C*.

**New since 2025-03-18.** RNA endonuclease activity (F, IBA); RNA catabolic process (P, IBA).

**To look for.** The supporting neighbors are SNAREs and a vacuolar sorting adaptor, while SGD's description and the two new phylogenetic (IBA) annotations point to a nuclease. The two pictures disagree and the transfer is the weaker one here: no single seed calls it, no other arm gives it, and it does not survive the widest-spread column dropout. Treat it as a low-confidence call. To check: the nuclease homology SGD cites and any vesicle trafficking phenotype.

## 4.6 *YER186C* ✗

**SGD.** Verified: "Protein of unknown function". Kuzmin et al. measured it in 11 array-role cells, 9 of them R2 training cells, and in 0 query-role cells.

**Prediction.** protein-RNA complex assembly (GO:0022618), $q = 1.2 \times 10^{-3}$: 12 of its 85 annotated neighbors bear the term against 2.41 expected by chance (124 bearers in the population); it has 135 neighbors in all and 4 transferred terms. GO defines the term: "The aggregation, arrangement and bonding together of proteins and RNA molecules to form a ribonucleoprotein complex."

**Supporting neighbors.** *RPS14B* (0.87), *YJU2* (0.75), *PRP28* (0.71), *RPL3* (0.68), *DBP10* (0.65), *CLF1* (0.65), *SYF1* (0.63), *PRP5* (0.63), *PRP19* (0.63), *RPL14A* (0.62), *RPL6A* (0.62), *RPF2* (0.61), with PCC. SGD describes the strongest three as *RPS14B*, "Protein component of the small (40S) ribosomal subunit"; *YJU2*, "Essential protein required for pre-mRNA splicing"; *PRP28*, "Pre-mRNA splicing factor".

**What carries the correlation.** Two query pairs carry 0.25 of each supporting PCC on average; the three strongest are NPP1-NPP2, GRX7-GRX6, ECM13-YJR115W. Of the 26 cells at the two strongest pairs (the gene and its supporting neighbors), 2 were measured, 2 of them R2 training cells. Its most negative pair-centered predictions: NPP1-NPP2 −0.156; CLA4-SKM1 −0.115; LSB1-PIN3 −0.104; its most positive: GRX7-GRX6 +0.116; MLP2-MLP1 +0.060.

**Robustness.** 0 of the three single-seed networks call the term (1 call it or a term within two steps); 0.08 of the supporting neighbors are kept in all three seeds by top-$k$ recovery; degenerate in seed: none. Other arms giving the same term: none; the term or one within two steps: none. The call survives the widest-spread column dropout: yes; random dropouts: 5 of 5.

**Baselines.** A share of 0.00 of the supporting neighbors are adjacent to it in an input graph. Union-graph guilt by association calls the term: no (within two steps: no); embedding guilt by association: no (within two steps: no). Its nearest genes in the learned embedding: *YDR509W*, *YPR064W*, *YMR141C*, *LCL2*, *YLR280C*.

**New since 2025-03-18.** none.

**To look for.** The supporting neighbors are ribosomal proteins, splicing factors and an rRNA maturation factor, a broad RNA-processing neighborhood of 135 genes. A neighborhood that large may reflect how little the model knows about the gene rather than its biology (hypothesis, untested). To check: splicing or ribosome biogenesis phenotypes of the deletion.

## 4.7 *YGL007C-A* ✗

**SGD.** Uncharacterized: "Putative protein of unknown function; identified by gene-trapping, microarray-based expression analysis, and genome-wide homology searching; deletion exhibits slow-growth phenotype; computationally predicted to have a role in cell budding". Kuzmin et al. measured it in 0 array-role cells, 0 of them R2 training cells, and in 0 query-role cells.

**Prediction.** rRNA processing (`GO:0006364`), $q = 2.8 \times 10^{-3}$: 23 of its 174 annotated neighbors bear the term against 8.54 expected by chance (215 bearers in the population); it has 223 neighbors in all and 22 transferred terms. GO defines the term: "Any process involved in the conversion of a primary ribosomal RNA (rRNA) transcript into one or more mature rRNA molecules."

**Supporting neighbors.** *RPS31* (0.93), *POP5* (0.93), *RPL30* (0.93), *RPS5* (0.93), *RRP36* (0.92), *RIO1* (0.92), *PWP2* (0.91), *KRI1* (0.91), *RRP7* (0.91), *POP3* (0.90), *RIO2* (0.89), *POP1* (0.89), *SNU13* (0.89), *POL5* (0.88), *SPB1* (0.88), *NUG1* (0.86), *IPI3* (0.85), *CSL4* (0.84), *BUD22* (0.83), *RRP17* (0.79), *UTP11* (0.69), *PAF1* (0.68), *IMP3* (0.61), with PCC. SGD describes the strongest three as *RPS31*, "Fusion protein cleaved to yield ribosomal protein S31 and ubiquitin"; *POP5*, "Subunit of both RNase MRP and nuclear RNase P"; *RPL30*, "Ribosomal 60S subunit protein L30".

**What carries the correlation.** Two query pairs carry 0.15 of each supporting PCC on average; the three strongest are GRX7-GRX6, ARE1-ARE2, YUR1-KTR2. Of the 48 cells at the two strongest pairs (the gene and its supporting neighbors), 0 were measured, 0 of them R2 training cells. Its most negative pair-centered predictions: ARE1-ARE2 −0.192; MLP2-MLP1 −0.160; STB6-STB2 −0.150; its most positive: YUR1-KTR2 +0.178; SWI5-ACE2 +0.141.

**Robustness.** 2 of the three single-seed networks call the term (2 call it or a term within two steps); 0.04 of the supporting neighbors are kept in all three seeds by top-$k$ recovery; degenerate in seed: `8fuwsk48`. Other arms giving the same term: none; the term or one within two steps: none. The call survives the widest-spread column dropout: yes; random dropouts: 5 of 5.

**Baselines.** A share of 0.00 of the supporting neighbors are adjacent to it in an input graph. Union-graph guilt by association calls the term: no (within two steps: no); embedding guilt by association: yes (within two steps: yes). Its nearest genes in the learned embedding: *SPC25*, *PET9*, *MYO1*, *YKL083W*, *TEN1*.

**New since 2025-03-18.** none.

**To look for.** The screen never measured *YGL007C-A*, so its profile is entirely inferred; it is degenerate in one seed and has 223 neighbors, most of them ribosome biogenesis genes, at higher PCCs than any other drawn candidate's neighbors. Hypothesis (untested): the profile is the model's low-information output mixed with one seed's unseen-gene profile, and the ribosome biogenesis neighborhood follows from that. The embedding route also gives the term, so the call may come from the learned gene representation rather than from triple interactions. SGD's description cites a slow-growth deletion phenotype and a predicted role in budding; check ribosome biogenesis readouts (polysome profiles, pre-rRNA processing) before reading the term as function.

## 4.8 *YLR001C* ✗

**SGD.** Verified: "Protein of unknown function; the authentic, non-tagged protein is detected in highly purified mitochondria in high-throughput studies; predicted to be palmitoylated". Kuzmin et al. measured it in 11 array-role cells, 8 of them R2 training cells, and in 0 query-role cells.

**Prediction.** dephosphorylation (`GO:0016311`), $q = 3.1 \times 10^{-3}$: 3 of its 13 annotated neighbors bear the term against 0.05 expected by chance (16 bearers in the population); it has 22 neighbors in all and 2 transferred terms. GO defines the term: "The process of removing one or more phosphoric (ester or anhydride) residues from a molecule."

**Supporting neighbors.** *LTP1* (0.67), *PTC7* (0.64), *YNL108C* (0.62), with PCC. SGD describes the strongest three as *LTP1*, "Protein phosphotyrosine phosphatase of unknown cellular role"; *PTC7*, "Type 2C serine/threonine protein phosphatase (PP2C)"; *YNL108C*, "Protein phosphatase".

**What carries the correlation.** Two query pairs carry 0.21 of each supporting PCC on average; the three strongest are PCL6-PCL7, CIK1-VIK1, PMD1-MDS3. Of the 8 cells at the two strongest pairs (the gene and its supporting neighbors), 4 were measured, 4 of them R2 training cells. Its most negative pair-centered predictions: PCL6-PCL7 −0.057; TOS8-CUP9 −0.035; GPB2-GPB1 −0.043, measured −0.047 (train); its most positive: BUG1-RET2 −0.048; MTC1-MDY2 −0.042.

**Robustness.** 0 of the three single-seed networks call the term (0 call it or a term within two steps); 0.00 of the supporting neighbors are kept in all three seeds by top-$k$ recovery; degenerate in seed: none. Other arms giving the same term: none; the term or one within two steps: Q1. The call survives the widest-spread column dropout: no; random dropouts: 5 of 5.

**Baselines.** A share of 0.00 of the supporting neighbors are adjacent to it in an input graph. Union-graph guilt by association calls the term: no (within two steps: no); embedding guilt by association: no (within two steps: no). Its nearest genes in the learned embedding: *YBR051W*, *YLR415C*, *YIL059C*, *YGL165C*, *YGL034C*.

**New since 2025-03-18.** none.

**To look for.** Three phosphatases support the term, one of them (*PTC7*) described by SGD as a PP2C. SGD places *YLR001C* in purified mitochondria in high-throughput data. No single seed calls the term and it does not survive the widest-spread dropout; Q1 gives a term within two steps. To check: whether the protein carries a phosphatase domain, and mitochondrial phosphatase substrates.

## 4.9 *PIR5* (YJL160C) ✗

**SGD.** Verified: "Protein of unknown function; member of the PIR (Proteins with Internal Repeats) family of cell wall proteins; SWAT-GFP fusion protein localizes to the endoplasmic reticulum and vacuole, and mCherry fusion localizes to the vacuole; non-essential gene that is required for sporulation; mRNA is weakly cell cycle regulated, peaking in mitosis; YJL160C has a paralog, PIR1, that arose from the whole genome duplication". Kuzmin et al. measured it in 9 array-role cells, 7 of them R2 training cells, and in 0 query-role cells.

**Prediction.** protein targeting to vacuole (GO:0006623), $q = 5.8 \times 10^{-3}$: 4 of its 9 annotated neighbors bear the term against 0.19 expected by chance (94 bearers in the population); it has 16 neighbors in all and 12 transferred terms. GO defines the term: "The process of directing proteins towards the vacuole, usually using signals contained within the protein."

**Supporting neighbors.** *ATG20* (0.68), *APS3* (0.67), *HSE1* (0.66), *APM3* (0.61), with PCC. SGD describes the strongest three as *ATG20*, "Sorting nexin family member"; *APS3*, "Small subunit of the clathrin-associated adaptor complex AP-3"; *HSE1*, "Subunit of the endosomal Vps27p-Hse1p complex".

**What carries the correlation.** Two query pairs carry 0.38 of each supporting PCC on average; the three strongest are CLN1-CLN2, NPP1-NPP2, YEH1-YEH2. Of the 10 cells at the two strongest pairs (the gene and its supporting neighbors), 6 were measured, 5 of them R2 training cells. Its most negative pair-centered predictions: CLN1-CLN2 −0.168; HBS1-SKI7 −0.095, measured −0.111 (train); MIY3-MIY2 −0.061; its most positive: NPP1-NPP2 +0.125; ERV29-ERV25 −0.078.

**Robustness.** 0 of the three single-seed networks call the term (0 call it or a term within two steps); 0.00 of the supporting neighbors are kept in all three seeds by top-$k$ recovery; degenerate in seed: none. Other arms giving the same term: none; the term or one within two steps: none. The call survives the widest-spread column dropout: no; random dropouts: 4 of 5.

**Baselines.** A share of 0.00 of the supporting neighbors are adjacent to it in an input graph. Union-graph guilt by association calls the term: no (within two steps: no); embedding guilt by association: no (within two steps: no). Its nearest genes in the learned embedding: *YDR262W*, *YGR164W*, *YJL067W*, *PUG1*, *YLR252W*.

**New since 2025-03-18.** none.

**To look for.** The supporting neighbors are two AP-3 adaptor subunits, a sorting nexin and a subunit of the endosomal Vps27-Hse1 complex, and SGD's own description puts fluorescent fusions of *PIR5* at the endoplasmic reticulum and the vacuole. That agreement is consistent with the transfer but does not test it: SGD also places *PIR5* in the PIR family of cell wall proteins, so a vacuolar signal could be cargo in transit (hypothesis, untested). Hypothesis (untested): *PIR5* reaches the vacuole by the AP-3 route. To check: AP-3 dependence of its vacuolar localization.

## 4.10 *IRC13* (YOR235W) ✗

**SGD.** Uncharacterized: "Putative protein of unknown function; conserved across S. cerevisiae strains; null mutant displays increased levels of spontaneous Rad52 foci". Kuzmin et al. measured it in 9 array-role cells, 7 of them R2 training cells, and in 0 query-role cells.

**Prediction.** tRNA aminoacylation for protein translation (GO:0006418), $q = 6.1 \times 10^{-3}$: 6 of its 94 annotated neighbors bear the term against 0.62 expected by chance (29 bearers in the population); it has 126 neighbors in all and 9 transferred terms. GO defines the term: "The synthesis of aminoacyl tRNA by the formation of an ester bond between the 3'-hydroxyl group of the most 3' adenosine of the tRNA and the alpha carboxylic acid group of an amino acid, to be used in ribosome-mediated polypeptide synthesis."

**Supporting neighbors.** *MES1* (0.79), *GRS1* (0.74), *THS1* (0.72), *GUS1* (0.65), *FRS2* (0.63), *NAM2* (0.62), with PCC. SGD describes the strongest three as *MES1*, "Methionyl-tRNA synthetase"; *GRS1*, "Cytoplasmic and mitochondrial glycyl-tRNA synthase"; *THS1*, "Threonyl-tRNA synthetase involved in tRNA aminoacylation".

**What carries the correlation.** Two query pairs carry 0.29 of each supporting PCC on average; the three strongest are MLP2-MLP1, TOS8-CUP9, GRX7-GRX6. Of the 14 cells at the two strongest pairs (the gene and its supporting neighbors), 4 were measured, 4 of them R2 training cells. Its most negative pair-centered predictions: TOS8-CUP9 −0.129; ERV29-ERV25 −0.242; PCL6-PCL7 −0.044; its most positive: MLP2-MLP1 +0.234; GRX7-GRX6 +0.200.

**Robustness.** 1 of the three single-seed networks call the term (1 call it or a term within two steps); 0.17 of the supporting neighbors are kept in all three seeds by top-$k$ recovery; degenerate in seed: none. Other arms giving the same term: none; the term or one within two steps: none. The call survives the widest-spread column dropout: no; random dropouts: 5 of 5.

**Baselines.** A share of 0.00 of the supporting neighbors are adjacent to it in an input graph. Union-graph guilt by association calls the term: no (within two steps: no); embedding guilt by association: no (within two steps: yes). Its nearest genes in the learned embedding: *PTH1*, *BRP1*, *RPS29A*, *RPO31*, *RPP0*.

**New since 2025-03-18.** none.

**To look for.** Six aminoacyl-tRNA synthetases support the term, among 126 neighbors. SGD reports increased spontaneous Rad52 foci in the null mutant. To check: translation-fidelity or amino-acid starvation phenotypes, and whether the protein co-purifies with the synthetases.

## 4.11 *YLR031W* ✗

**SGD.** Verified: "Protein of unknown function; S288C contains an in-frame stop codon between ORFs YLR030W and YLR031W; YLR031W has a paralog, YMR124W, that arose from the whole genome duplication". Kuzmin et al. measured it in 11 array-role cells, 10 of them R2 training cells, and in 1,042 query-role cells.

**Prediction.** monosaccharide metabolic process (GO:0005996), $q = 6.3 \times 10^{-3}$: 5 of its 37 annotated neighbors bear the term against 0.37 expected by chance (44 bearers in the population); it has 54 neighbors in all and 3 transferred terms. GO defines the term: "The chemical reactions and pathways involving monosaccharides, the simplest carbohydrates."

**Supporting neighbors.** *GAL3* (0.66), *YPR1* (0.62), *EMI2* (0.61), *GCY1* (0.60), *STD1* (0.60), with PCC. SGD describes the strongest three as *GAL3*, "Transcriptional regulator"; *YPR1*, "NADPH-dependent aldo-keto reductase"; *EMI2*, "Hexokinase expressed during growth with low glucose levels".

**What carries the correlation.** Two query pairs carry 0.18 of each supporting PCC on average; the three strongest are GRX7-GRX6, BUG1-RET2, MLP2-MLP1. Of the 12 cells at the two strongest pairs (the gene and its supporting neighbors), 0 were measured, 0 of them R2 training cells. Its most negative pair-centered predictions: GRX7-GRX6 −0.052; SWI5-ACE2 −0.028; YEA6-YIA6 −0.005; its most positive: BUG1-RET2 −0.022; MTC1-MDY2 −0.036.

**Robustness.** 1 of the three single-seed networks call the term (2 call it or a term within two steps); 0.00 of the supporting neighbors are kept in all three seeds by top-$k$ recovery; degenerate in seed: none. Other arms giving the same term: R1; the term or one within two steps: R1. The call survives the widest-spread column dropout: no; random dropouts: 4 of 5. In Tier B it has 24 neighbors, 18 of them shared with Tier A; its Tier B AUC for the term is 0.56, above 53% of the decoys.

**Baselines.** A share of 0.00 of the supporting neighbors are adjacent to it in an input graph. Union-graph guilt by association calls the term: no (within two steps: no); embedding guilt by association: no (within two steps: no). Its nearest genes in the learned embedding: *PDR3*, *PCL9*, *HEH2*, *ALK1*, *AGA1*.

**New since 2025-03-18.** none.

**To look for.** The supporting neighbors are galactose and glucose regulators and carbohydrate enzymes. *YLR031W* is a query-pair gene in the screen (paired with *EPO1*), so it has many query-role cells but few array-role cells, and its profile is the array-role one. R1 gives the same term. To check: growth on galactose and low glucose, and its whole-genome-duplication paralog named by SGD.

## 4.12 *YDR249C* ✗

**SGD.** Verified: "Protein of unknown function". Kuzmin et al. measured it in 9 array-role cells, 7 of them R2 training cells, and in 0 query-role cells.

**Prediction.** endoplasmic reticulum to Golgi vesicle-mediated transport (GO:0006888), $q = 7.2 \times 10^{-3}$: 6 of its 50 annotated neighbors bear the term against 0.64 expected by chance (56 bearers in the population); it has 65 neighbors in all and 9 transferred terms. GO defines the term: "The directed movement of substances from the endoplasmic reticulum (ER) to the Golgi, mediated by COP II vesicles."

**Supporting neighbors.** *SVP26* (0.69), *BET3* (0.67), *MRS6* (0.64), *GRH1* (0.63), *YPT1* (0.62), *SLY1* (0.61), with PCC. SGD describes the strongest three as *SVP26*, "ER exit adaptor for Golgi mannosyltransferases"; *BET3*, "Core component of transport protein particle (TRAPP) complexes I-III"; *MRS6*, "Rab escort protein".

**What carries the correlation.** Two query pairs carry 0.44 of each supporting PCC on average; the three strongest are BUG1-RET2, MLP2-MLP1, ERV29-ERV25. Of the 14 cells at the two strongest pairs (the gene and its supporting neighbors), 7 were measured, 6 of them R2 training cells. Its most negative pair-centered predictions: ERV29-ERV25 −0.341; BUG1-RET2 −0.272; MLP2-MLP1 −0.158, measured −0.137 (train); its most positive: SWI5-ACE2 +0.081; CPR4-CPR8 +0.018.

**Robustness.** 1 of the three single-seed networks call the term (1 call it or a term within two steps); 0.00 of the supporting neighbors are kept in all three seeds by top-$k$ recovery; degenerate in seed: none. Other arms giving the same term: none; the term or one within two steps: none. The call survives the widest-spread column dropout: yes; random dropouts: 4 of 5.

**Baselines.** A share of 0.00 of the supporting neighbors are adjacent to it in an input graph. Union-graph guilt by association calls the term: no (within two steps: no); embedding guilt by association: no (within two steps: no). Its nearest genes in the learned embedding: *YOR325W*, *YHL041W*, *YKR012C*, *YOR186W*, *YLF2*.

**New since 2025-03-18.** none.

**To look for.** The supporting neighbors are early secretory pathway genes (TRAPP, a Rab escort protein, *YPT1*, *SLY1*, a GRASP homolog, an ER exit adaptor), and *BOP2* is itself among its neighbors. Hypothesis (untested): *YDR249C* and *BOP2* belong to one secretory module the model sees through the same few query pairs. To check: ER-to-Golgi transport readouts in the deletion.

## 4.13 *MRX1* (YER077C) ✗

**SGD.** Verified: "Protein that associates with mitochondrial ribosome; green fluorescent protein (GFP)-fusion protein localizes to the mitochondrion; null mutation results in a decrease in plasma membrane electron transport". Kuzmin et al. measured it in 0 array-role cells, 0 of them R2 training cells, and in 0 query-role cells.

**Prediction.** spindle pole body duplication (`GO:0030474`), $q = 9.2 \times 10^{-3}$: 5 of its 150 annotated neighbors bear the term against 0.45 expected by chance (13 bearers in the population); it has 182 neighbors in all and 17 transferred terms. GO defines the term: "Construction of a new spindle pole body."

**Supporting neighbors.** *NDC1* (0.66), *MPS2* (0.66), *KAR1* (0.64), *MPS3* (0.61), *CDC31* (0.61), with PCC. SGD describes the strongest three as *NDC1*, "Subunit of the transmembrane ring of the nuclear pore complex (NPC)"; *MPS2*, "Essential membrane protein localized at nuclear envelope and SPBs"; *KAR1*, "Protein involved in karyogamy and spindle pole body duplication".

**What carries the correlation.** Two query pairs carry 0.14 of each supporting PCC on average; the three strongest are STB6-STB2, LPX2-EIS1, ARE1-ARE2. Of the 12 cells at the two strongest pairs (the gene and its supporting neighbors), 0 were measured, 0 of them R2 training cells. Its most negative pair-centered predictions: STB6-STB2 −0.131; LPX2-EIS1 −0.102; ARE1-ARE2 −0.104; its most positive: CLN1-CLN2 +0.087; GPB2-GPB1 +0.092.

**Robustness.** 0 of the three single-seed networks call the term (0 call it or a term within two steps); 0.00 of the supporting neighbors are kept in all three seeds by top-$k$ recovery; degenerate in seed: `8fuwsk48`. Other arms giving the same term: Q1; the term or one within two steps: Q1. The call survives the widest-spread column dropout: yes; random dropouts: 5 of 5.

**Baselines.** A share of 0.00 of the supporting neighbors are adjacent to it in an input graph. Union-graph guilt by association calls the term: no (within two steps: no); embedding guilt by association: yes (within two steps: yes). Its nearest genes in the learned embedding: *ERG24*, *HYP2*, *RRP17*, *VMA10*, *VMA7*.

**New since 2025-03-18.** none.

**To look for.** The screen never measured *MRX1*; its profile is degenerate in one seed and has 182 neighbors. SGD describes a mitochondrial ribosome-associated protein, which the spindle pole body term contradicts. Q1 and the embedding route give the same term, which points to the learned representation rather than to triple interactions (hypothesis, untested). Low confidence.

## 4.14 *YBR220C* ✗

**SGD.** Verified: "Protein of unknown function; YBR220C is not an essential gene". Kuzmin et al. measured it in 10 array-role cells, 9 of them R2 training cells, and in 0 query-role cells.

**Prediction.** polyphosphate metabolic process (GO:0006797), $q = 1.1 \times 10^{-2}$: 2 of its 4 annotated neighbors bear the term against 0.01 expected by chance (14 bearers in the population); it has 6 neighbors in all and 1 transferred terms. GO defines the term: "The chemical reactions and pathways involving a polyphosphate, the anion or salt of polyphosphoric acid."

**Supporting neighbors.** *VTC5* (0.66), *VTC3* (0.61), with PCC. SGD describes the strongest three as *VTC5*, "Novel subunit of the vacuolar transporter chaperone complex"; *VTC3*, "Regulatory subunit of vacuolar transporter chaperone (VTC) complex".

**What carries the correlation.** Two query pairs carry 0.27 of each supporting PCC on average; the three strongest are MSA2-MSA1, ERV29-ERV25, MIY3-MIY2. Of the 6 cells at the two strongest pairs (the gene and its supporting neighbors), 3 were measured, 3 of them R2 training cells. Its most negative pair-centered predictions: USO1-SEC31 −0.149; COX5B-COX5A −0.045; CHS6-BCH2 −0.049, measured −0.056 (train); its most positive: ERV29-ERV25 −0.057; MSA2-MSA1 +0.093.

**Robustness.** 0 of the three single-seed networks call the term (0 call it or a term within two steps); 0.00 of the supporting neighbors are kept in all three seeds by top-$k$ recovery; degenerate in seed: none. Other arms giving the same term: none; the term or one within two steps: R1, Q1. The call survives the widest-spread column dropout: yes; random dropouts: 4 of 5.

**Baselines.** A share of 0.00 of the supporting neighbors are adjacent to it in an input graph. Union-graph guilt by association calls the term: no (within two steps: yes); embedding guilt by association: no (within two steps: no). Its nearest genes in the learned embedding: *YIL163C*, *CSS2*, *PTK1*, *YOR379C*, *YPR147C*.

**New since 2025-03-18.** endoplasmic reticulum membrane (C, IBA); acetyl-CoA transmembrane transporter activity (F, IBA); acetyl-CoA transmembrane transport (P, IBA).

**To look for.** Two subunits of the vacuolar transporter chaperone complex, both described by SGD as involved in vacuolar polyphosphate accumulation, are its only supporting neighbors among 6. SGD's new phylogenetic annotations since 2025-03-18 say acetyl-CoA transmembrane transport at the ER membrane, which is not within two steps of the transferred term. R1, Q1 and the union of the input graphs reach a term within two steps. To check: polyphosphate levels in the deletion, and the acetyl-CoA transporter family the IBA annotation implies.

## 4.15 *YDL009C* ✗

**SGD.** Uncharacterized: "Protein of unknown function; mRNA identified as translated by ribosome profiling data; SWAT-GFP and mCherry fusion proteins localize to the cytosol; partially overlaps the verified ORF YDL010W; YDL009C is not an essential gene". Kuzmin et al. measured it in 0 array-role cells, 0 of them R2 training cells, and in 0 query-role cells.

**Prediction.** transcription initiation at RNA polymerase II promoter (`GO:0006367`), $q = 1.2 \times 10^{-2}$: 5 of its 161 annotated neighbors bear the term against 0.48 expected by chance (13 bearers in the population); it has 205 neighbors in all and 23 transferred terms. GO defines the term: "A transcription initiation process that takes place at a RNA polymerase II gene promoter."

**Supporting neighbors.** *SRB4* (0.78), *SSU72* (0.76), *RPB4* (0.73), *RPB9* (0.62), *TOA2* (0.62), with PCC. SGD describes the strongest three as *SRB4*, "Subunit of the RNA polymerase II mediator complex"; *SSU72*, "Phosphatase and transcription/RNA-processing factor"; *RPB4*, "RNA polymerase II subunit B32".

**What carries the correlation.** Two query pairs carry 0.18 of each supporting PCC on average; the three strongest are SWI5-ACE2, GRX7-GRX6, ARE1-ARE2. Of the 12 cells at the two strongest pairs (the gene and its supporting neighbors), 0 were measured, 0 of them R2 training cells. Its most negative pair-centered predictions: ERV29-ERV25 −0.251; SED1-SPI1 −0.091; MLP2-MLP1 −0.090; its most positive: SWI5-ACE2 +0.121; GRX7-GRX6 +0.074.

**Robustness.** 0 of the three single-seed networks call the term (1 call it or a term within two steps); 0.00 of the supporting neighbors are kept in all three seeds by top-$k$ recovery; degenerate in seed: `8fuwsk48`. Other arms giving the same term: none; the term or one within two steps: none. The call survives the widest-spread column dropout: no; random dropouts: 5 of 5.

**Baselines.** A share of 0.00 of the supporting neighbors are adjacent to it in an input graph. Union-graph guilt by association calls the term: no (within two steps: no); embedding guilt by association: no (within two steps: no). Its nearest genes in the learned embedding: *YJL015C*, *CWC25*, *NAR1*, *YAL034C-B*, *YFH1*.

**New since 2025-03-18.** none.

**To look for.** The screen never measured *YDL009C*; its profile is degenerate in one seed and has 205 neighbors, the supporting ones RNA polymerase II, Mediator and TFIIA subunits. SGD notes that the ORF partially overlaps the verified ORF *YDL010W*, so a deletion of *YDL009C* may disturb its neighbor; any phenotype has to be read with that in mind. Low confidence.

### 4.16 *YPL245W* ✗

**SGD.** Verified: "Protein of unknown function; green fluorescent protein (GFP)-fusion protein localizes to both the nucleus and the cytoplasm". Kuzmin et al. measured it in 10 array-role cells, 8 of them R2 training cells, and in 497 query-role cells.

**Prediction.** homologous chromosome pairing at meiosis (`GO:0007129`), $q = 1.2 \times 10^{-2}$: 2 of its 5 annotated neighbors bear the term against 0.01 expected by chance (12 bearers in the population); it has 10 neighbors in all and 4 transferred terms. GO defines the term: "The meiotic cell cycle process where side by side pairing and physical juxtaposition of homologous chromosomes is created during meiotic prophase."

**Supporting neighbors.** *ZIP2* (0.62), *SPO11* (0.61), with PCC. SGD describes the strongest three as *ZIP2*, "Meiosis-specific protein"; *SPO11*, "Meiosis-specific protein required for meiotic recombination".

**What carries the correlation.** Two query pairs carry 0.24 of each supporting PCC on average; the three strongest are ERV29-ERV25, PCL6-PCL7, CPR4-CPR8. Of the 6 cells at the two strongest pairs (the gene and its supporting neighbors), 0 were measured, 0 of them R2 training cells. Its most negative pair-centered predictions: ASM4-NUP53 −0.087; PCL6-PCL7 −0.048; TOS8-CUP9 −0.024; its most positive: MTC1-MDY2 −0.041; SDC1-SET1 −0.039.

**Robustness.** 0 of the three single-seed networks call the term (0 call it or a term within two steps); 0.00 of the supporting neighbors are kept in all three seeds by top-$k$ recovery; degenerate in seed: none. Other arms giving the same term: Q1, Q2; the term or one within two steps: Q1, Q2. The call survives the widest-spread column dropout: yes; random dropouts: 5 of 5. In Tier B it has 294 neighbors, 9 of them shared with Tier A; its Tier B AUC for the term is 0.64, above 88% of the decoys.

**Baselines.** A share of 0.00 of the supporting neighbors are adjacent to it in an input graph. Union-graph guilt by association calls the term: no (within two steps: no); embedding guilt by association: yes (within two steps: yes). Its nearest genes in the learned embedding: *YIL046W-A*, *HGH1*, *SPO11*, *ZIP2*, *YDR169C-A*.

**New since 2025-03-18.** none.

**To look for.** Two meiosis-specific proteins support the term, and both are also among its five nearest genes in the learned embedding; Q1, Q2 and the embedding route give the same term. The call is therefore one the model's gene representation already carries (hypothesis, untested: inherited from the embedding rather than learned from triple interactions). To check: sporulation efficiency and meiotic recombination in the deletion.

## 5 Other routes to attribution ✗

The routes below were run over all 93 validation candidates, with the 16 prospective genes as the set whose curation outcome is known. They are ordered by the question each answers: has SGD curated anything since the frozen GAF, does the call hold in each checkpoint, which query pairs carry it, do other models give it, and do the model's own inputs already imply it. Terms are defined in Tables 1 and 4.

### 5.1 Annotations SGD added after the frozen GAF ✗

Between the frozen GAF (2025-03-18) and SGD's GAF of 2026-10-20, 11 of the 93 candidates gained at least one annotation outside ND (25 new gene, term and code rows), and every one is computational: 5 genes by phylogenetic inference (IBA, 14 rows) and 8 by electronic inference (IEA, 11 rows). No candidate gained an experimental, high-throughput or genetic-interaction annotation in any aspect; for biological process that holds by construction, since a candidate has no experimental BP term in either file, and for function and component it is a finding. By aspect, 9 genes gained a process term, 9 a function term and 4 a component term. None of the new annotations is within two `is_a` steps of a term transferred to the same gene (0 genes), none within two steps over `is_a` and `part_of` (0), and none through a GO id in an annotation extension (0). In the other direction, neither GAF annotates a candidate to any of its 374 transferred terms or to a descendant of one, under any code (0 in the frozen GAF, 0 in the 2026-10-20 GAF), and the 2026-10-20 GAF holds no term within two steps of one (0). Among the drawn candidates, 2 have new annotations, *LCL3* and *YBR220C*, all phylogenetic (Figure 14).

The prospective genes are the comparison: all 16 gained annotations (44 rows), 16 of them experimental (IDA for 9, IMP for 13), 3 genetic interaction. Only 5 of the 16 had any transferred term, and no new annotation of any code sits within two steps of one (0 genes; Table 6).

The two files come from different producers, a GO Consortium release and SGD's own file, so computational rows also change for reasons other than new knowledge. Genome-wide, 3,996 IEA rows are new and 13,881 gone of 32,348; IBA 2,235 new and 1,545 gone of 15,140; RCA 2,159 new; against IDA 550 new of 14,313 and IMP 486 new of 11,762.

**What it adds.** Whether curation has reached any transfer in the 19 months since the frozen file: it has not, for any candidate, in any aspect or code.

**What it cannot show.** A transfer without new curation is untested, not wrong. The new computational annotations are themselves inferences from sequence and phylogeny, and SGD's GAF lags the literature, so the literature search the dossiers prepare remains the test.

**Table 6:** The 16 prospective genes: the experimental or genetic-interaction BP terms SGD added after the frozen GAF (verbatim GO names), the number of R2 transferred terms, the best transferred term, and the gene's measured array-role cells. None of the transferred terms is within two `is_a` steps of a new term.

| Gene | New experimental BP terms | Terms | Best transferred term | Cells |
|---|---|---|---|---|
| *OCA5* | diphosphoinositol polyphosphate metabolic process | 0 | – | 307 |
| *YBR062C* | IRES-dependent translational initiation of linear mRNA | 3 | amino acid transport | 294 |
| *IAT4* | melatonin biosynthetic process | 0 | – | 281 |
| *HOI1* | protein targeting to membrane; sterol homeostasis | 0 | – | 11 |
| *ALE2* | phosphatidylethanolamine acyl-chain remodeling | 0 | – | 11 |
| *ACL1* | ribosomal large subunit biogenesis; ribosomal protein import into nucleus | 1 | nucleotide-excision repair | 10 |
| *RTS3* | cellular response to nitrogen starvation; regulation of TORC1 signaling | 0 | – | 10 |
| *YPL062W* | terpenoid biosynthetic process | 0 | – | 10 |
| *PUT7* | regulation of proline catabolic process to glutamate | 0 | – | 9 |
| *FMP27* | sterol homeostasis | 1 | nonfunctional rRNA decay | 9 |
| *CIM1* | negative regulation of mitochondrial DNA metabolic process | 0 | – | 9 |
| *FMP10* | mitochondrial respiratory chain complex III assembly | 0 | – | 11 |
| *LDO45* | lipophagy | 1 | postreplication repair | 10 |
| *SNI445* | cytosolic ribosome assembly; rRNA acetylation involved in maturation of SSU-rRNA | 2 | cellular hyperosmotic response | 10 |
| *MRX4* | regulation of translational initiation | 0 | – | 10 |
| *YLR125W* | sulfur compound metabolic process | 0 | – | 9 |

## 5.2 Seed stability ✗

Each R2 checkpoint's own Tier A file, put through the same rule, gives a sparser network than the seed mean: 98, 209 and 544 ring 2 calls (`8fuwsk48`, `srl1xzdf`, `yb4gjh51`) against 462 calls on 107 genes for the seed mean, and more degenerate profiles (584, 566, 457 against 416). The best term of 0 candidates is called by all three single-seed networks; 4 are called by two, 16 by one and 73 by none, and 28 have the term or one within two steps called in at least one seed. On average 0.01 of a candidate's supporting neighbors stay above 0.6 in all three seeds and 0.04 are kept by top-$k$ recovery; the neighbor sets overlap the seed mean's with a mean Jaccard of 0.13 at 0.6 and 0.18 by top-$k$. 5 candidates are degenerate in at least one seed, 3 of them drawn.

**What it adds.** Whether a call reproduces in each checkpoint or exists only in the seed mean. In R2 it is almost always the latter.

**What it cannot show.** Whether the seed mean is right: averaging removes seed noise, so a call only the mean makes may be a real signal no single seed resolves (hypothesis, untested). The three seeds share architecture, data and split, so their agreement would not have been independent evidence either.

## 5.3 Which query pairs carry the correlations ✗

A candidate's two strongest query pairs carry a median 0.27 of each supporting PCC, against a median of 0.25 over 20,000 edges above 0.6 drawn from the whole network (90th percentile 0.55; 8 candidates lie above it). Concentration is therefore typical of the network, and the strongest pairs are the same ones across candidates: counting how many of the 93 candidates have a pair among their three strongest gives GRX7-GRX6 (31), NPP1-NPP2 (26), MLP2-MLP1 (21), PCL6-PCL7 (19), TOS8-CUP9 (15), and these five rank 1, 16, 2, 4, 7 of 420 in the spread of their pair-centered predictions over genes. Of the 862 cells at the candidates' two strongest pairs (each candidate and its supporting neighbors), 241 (28.0%) were measured by the screen, 211 of them R2 training cells; the rest are predictions.

Removing the 10 pairs with the widest spread (GRX7-GRX6, MLP2-MLP1, ERV29-ERV25, PCL6-PCL7, SWI5-ACE2, CLN1-CLN2, TOS8-CUP9, KRE2-KTR6, CIK1-VIK1, CPR4-CPR8) and rebuilding the network keeps the best term for 21 of the 93 candidates (26 keep it or a term within two steps), against 80.4 on average when 10 random pairs are removed (5 draws). Among the drawn candidates the counts are 8 and 13.2.

**What it adds.** Where the signal lives: most transfers depend on the few query pairs over which the model's predictions vary most, and 28.0% of the cells at the strongest pairs are measured, almost all of them training data. Those pairs are where a wet-lab trigenic test of a candidate is most direct.

**What it cannot show.** Whether those pairs reflect biology or the model. The screen itself has query pairs that interact with many genes, and a module interacting with one such pair is a legitimate signal (hypothesis, untested).

### 5.4 Agreement with the other arms ✗

The same rule on the Tier A seed means of R1, Q1 and Q2 (read from each arm's own transfer table) gives a candidate its R2 best term exactly for 7, 6 and 5 of the 93 candidates, and the term or one within two `is_a` steps for 11, 16 and 14; 29 candidates have such a match in at least one arm. A match within two steps is easier for an arm that calls many terms (median per candidate: R1 6, Q1 58, Q2 21), so each count is set against the matches expected by chance, the sum over candidates of the share of all gold terms within two steps of that arm's called set: 3.5 for R1, 12.8 for Q1 and 7.5 for Q2. Among the drawn candidates the within-two counts are 4, 4 and 1; for the 5 prospective genes with a transfer, none.

**What it adds.** R1 is a different architecture on the same random split, and the Q arms never saw 89 of the query pairs. R1's agreement is well above its chance expectation, Q2's above it, Q1's close to it.

**What it cannot show.** Correctness: every arm trains on the same Kuzmin cells, so a bias in those cells is shared.

### 5.5 The model's input graphs ✗

Across the 93 candidates a share of 0.03 of the best term's supporting neighbors are adjacent to the candidate in any of the nine input graphs; 8 candidates have at least one adjacent supporting neighbor and 1 has all of them. Guilt by association on the union graph, under the same rule and one Benjamini-Hochberg correction over the ring 2 tests, makes 32,028 calls but gives the best term to 5 candidates (12 the term or one within two steps); no single graph gives it to more (5 candidates over all nine).

**What it adds.** The plan's baseline (F7 item 6, G9): a transfer the inputs already imply is not new information, and for most candidates the inputs do not imply it.

**What it cannot show.** That the model's own transfers are right: they can come from the training cells or from structure the encoder learned, which the next route examines.

### 5.6 The learned gene embedding ✗

Over 20,000 sampled gene pairs the profile PCC and the embedding cosine correlate at 0.43, so the predicted profiles carry the encoder's gene representation in part; the cosine itself agrees between two seeds at 0.58. Guilt by association on each gene's $k$ nearest genes in the embedding, $k$ matched to its profile degree, makes 667 ring 2 calls on 115 genes and gives the best term to 6 of the 93 candidates (10 within two steps); on average 0.33 of a candidate's 20 nearest embedding genes are also its profile neighbors.

**What it adds.** It separates calls the representation already encodes (among the drawn, *YGL007C-A*, *MRX1* and *YPL245W*) from calls that appear only through predicted interactions.

**What it cannot show.** Independence: the embedding is learned from the same training cells and input graphs as the profiles.

### 5.7 Tier B module-internal columns ✗

Every gene's profile over the 5,651 Tier B core columns (seed mean of the three R2 files, the same preprocessing and threshold) gives 392 degenerate profiles, all of them degenerate in Tier A as well, and 91,015 edges; over gene pairs the Tier B PCC correlates with the Tier A PCC at 0.45, and the ring 2 rule makes 138 calls on 54 genes. For the 18

candidates in the 165-gene panel, the Tier A and Tier B neighbor sets overlap with a median Jaccard of 0.06; Tier B calls none of their best terms (0) and gives a term within two steps of an R2 term to 2; 2 candidates have no Tier B neighbor. Over the whole PCC row, the best term's bearers rank above non-bearers with a median AUC of 0.66 in Tier A and 0.62 in Tier B; the Tier B AUC exceeds 90% of the 100 degree-matched decoys for 6 candidates.

**What it adds.** New columns from the same model: a bearer AUC above the decoys in Tier B is module signal outside the 420 measured query pairs.

**What it cannot show.** Independence, since the checkpoints are the same; and an unbiased probe, since the panel was built around the prospective genes' term bearers and the dense uncharacterized genes (plan G9 keeps Tier B descriptive). The candidates in the panel are there as query-pair genes, so their own pairs are among the core columns and absent from their own profiles.

## 5.8 The drawn candidates across the routes ✗

Read across Table 5, *BOP2* and *MLO50* carry the most support among the drawn candidates: the two lowest $q$, a call in at least one single-seed network, the same term from R1, survival of the widest-spread column dropout, and no call from the input graphs or the learned embedding. Both also rest on the most concentrated correlations, about half of each supporting PCC in two query pairs, and both share a supporting neighbor (*ARL1*) and driving pairs, so they are one Golgi trafficking module seen twice rather than two independent calls (hypothesis, untested). *YGL007C-A*, *MRX1* and *YPL245W* are calls the learned embedding already makes, the first two for genes the screen never measured. *LCL3*, *YLR001C*, *PIR5* and *YDL009C* have no single-seed call, no other arm giving the term and no survival of the widest-spread dropout, which makes them the weakest. None of the 14 has a curated annotation that tests its transfer.

SRC: `candidate_dossiers_045.py, plot_since_annotated`

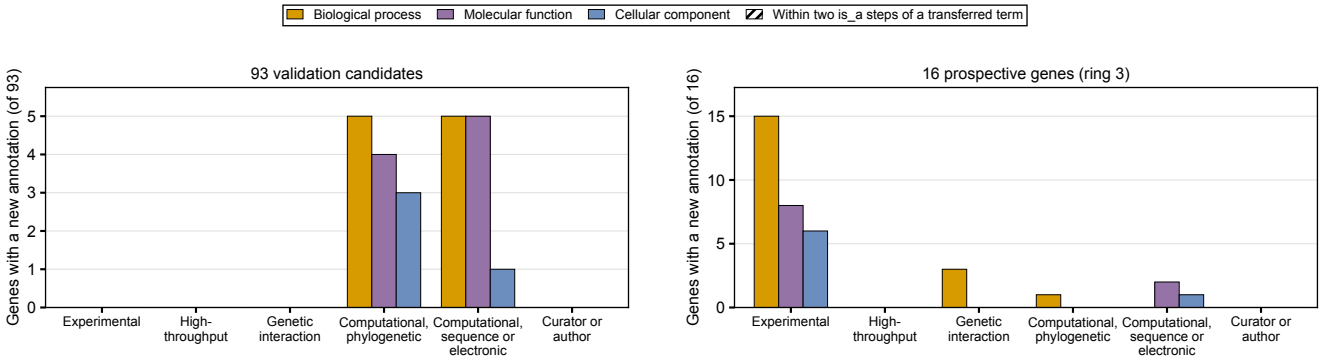



**Figure 14: No candidate gained an experimental annotation after the frozen GAF, and no new annotation of any code lies within two steps of a transferred term.** Genes with at least one new annotation (SGD GAF of 2026-10-20 against the frozen GAF of 2025-03-18; gene, term and code; ND excluded), by evidence class and GO aspect: orange biological process, purple molecular function, blue cellular component. A hatched overlay would mark genes with a new annotation within two `is_a` steps of a transferred term; there are none. **Left:** the 93 validation candidates, 11 with a new annotation, all computational. **Right:** the 16 prospective genes, all with new experimental or genetic-interaction annotations by definition, 5 of them with a transferred term. **Status: complete** for the two GAFs and the R2 transfers; the files come from different producers, so computational classes include pipeline differences.

SRC: `candidate_dossiers_045.py, plot_seed_concordance`

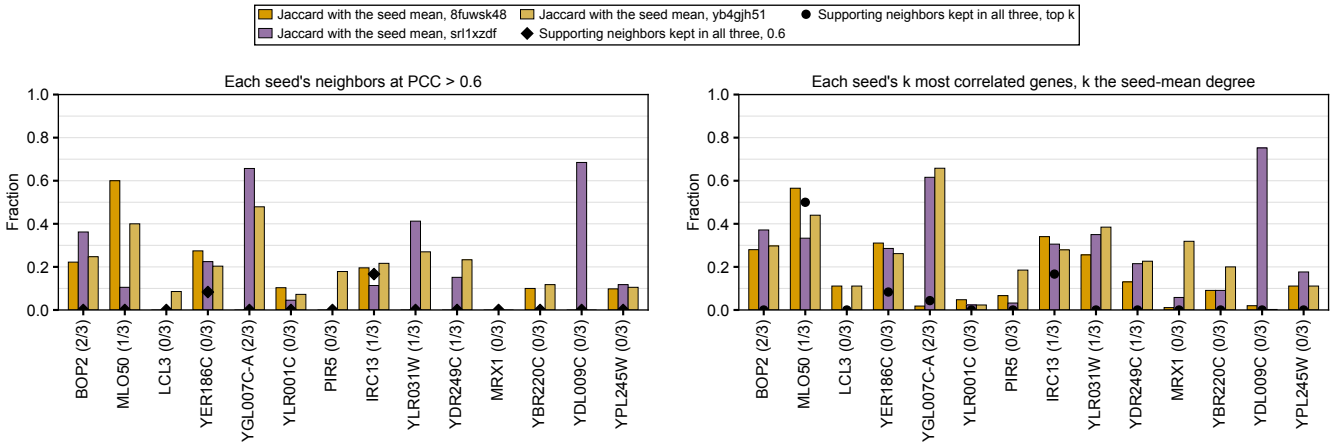



**Figure 15: The drawn candidates' neighborhoods change from seed to seed, and their supporting neighbors rarely survive in all three.** For each drawn candidate, the Jaccard index between one seed's neighbor set and the seed mean's (orange, purple and yellow bars for `8fuwsk48`, `srl1xzdf` and `yb4gjh51`), and the fraction of the best term's supporting neighbors kept in all three seeds (black marks); the count after each gene is the number of single-seed networks that call the term. **Left:** neighbor sets at PCC $> 0.6$; a single seed's PCCs run lower, so its sets are smaller. **Right:** each seed's $k$ most correlated genes, $k$ the seed-mean degree, which removes that size effect. **Status: complete for the three R2 Tier A files**; the Jaccard index of an empty set with a nonempty one is zero.

SRC: `candidate_dossiers_045.py, plot_tier_b`

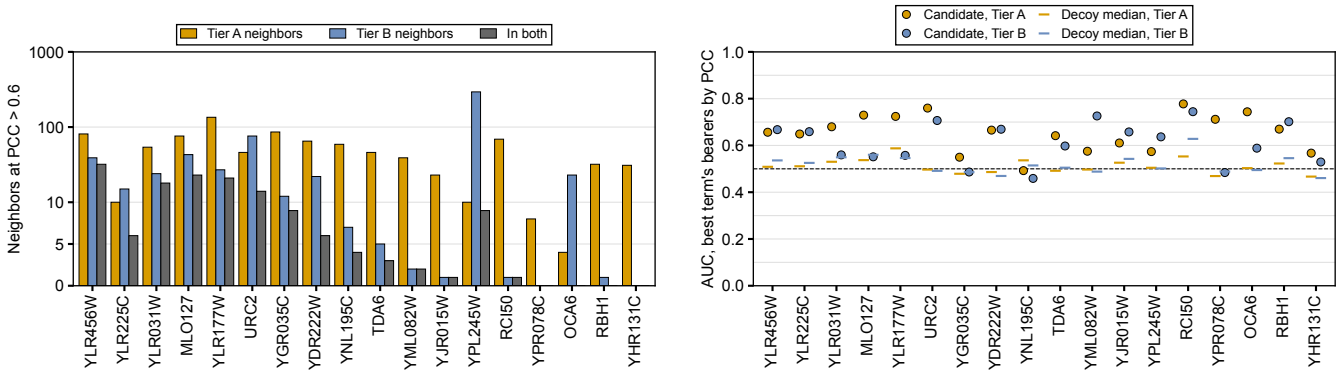



**Figure 16: Tier B does not reproduce the candidates' Tier A neighborhoods, but for several the best term's bearers still rank high in the Tier B row.** The 18 candidates in the 165-gene Tier B panel, ordered by the Jaccard index of their two neighbor sets. **Left:** neighbors at PCC $> 0.6$ over the 420 Tier A columns (orange), over the 5,651 Tier B core columns (blue) and in both (gray), symmetric-log axis. **Right:** the AUC with which the candidate's PCC row ranks its best term's bearers above the other annotated genes, Tier A (orange) and Tier B (blue), with the median AUC of 100 degree-matched decoys (bars); dashed line, no separation. The Tier A AUC is not independent of the call, which Tier A neighbors chose. **Status: complete for the 19 Tier B core blocks of the three R2 checkpoints**; descriptive, the panel favors the prospective genes' modules; the Tier B extension blocks are not used.